

VOPĚNKA'S PRINCIPLE WITHOUT CHOICE: PRESERVATION UNDER SYMMETRIC EXTENSIONS

TOM DE GROOT AND WOJCIECH ALEKSANDER WOŁOSZYN

ABSTRACT. We prove that every set-sized symmetric extension of a model of $\mathbf{ZF} + \mathbf{VP}$ again satisfies $\mathbf{ZF} + \mathbf{VP}$, where $\mathbf{VP}$ is formulated for arbitrary set-sized languages. For each standard $n \geq 1$, the same construction preserves $\mathbf{VP}(\Pi_n)$, allowing set parameters and arbitrary set-sized languages. The theories $\mathbf{ZF} + \mathbf{VP}$ and $\mathbf{ZFC} + \mathbf{VP}$ are equiconsistent, and $\mathbf{ZF} + \mathbf{VP}$ proves a proper class of Löwenheim–Skolem cardinals. Relative to $\text{Con}(\mathbf{ZF} + \mathbf{VP})$, the axiom $\mathbf{DC}$ is independent of $\mathbf{ZF} + \mathbf{VP} + \neg \mathbf{AC}$, and there are Feferman–Lévy and full Solovay models satisfying $\mathbf{VP}$. It remains open whether every countable model of $\mathbf{ZF} + \mathbf{VP}$ has a class-generic extension satisfying $\mathbf{ZFC} + \mathbf{VP}$.

1. INTRODUCTION

Vopěnka's principle asserts that every proper class of structures in a common set-sized language contains distinct members M and N with an elementary embedding $j : M \rightarrow N$. Throughout, $\mathbf{VP}$ denotes the corresponding first-order scheme for classes definable with set parameters. For each standard $n \geq 1$, $\mathbf{VP}(\Pi_n)$ restricts the defining formula to Π_n .

Over $\mathbf{ZFC}$, Bagaria showed that $\mathbf{VP}$ is equivalent to the existence of a $C^{(n)}$ -extendible cardinal for every n [1]. Far less has been published about $\mathbf{VP}$ without Choice. Tzouvaras proved that a general downward-absoluteness principle for $\mathbf{VP}$ implies $\mathbf{AC}$ and asked whether $\mathbf{AC}$ is independent of $\mathbf{ZF} + \mathbf{VP}$ [20, Question 4.1]. The following two theorems answer this question.

Theorem A. *Every set-sized symmetric extension of a model of $\mathbf{ZF} + \mathbf{VP}$ satisfies $\mathbf{ZF} + \mathbf{VP}$. For every standard $n \geq 1$, every set-sized symmetric extension of a model of $\mathbf{ZF} + \mathbf{VP}(\Pi_n)$ satisfies $\mathbf{ZF} + \mathbf{VP}(\Pi_n)$.*

Theorem B. *The theories $\mathbf{ZF} + \mathbf{VP}$ and $\mathbf{ZFC} + \mathbf{VP}$ are equiconsistent.*

Applied to the basic Cohen symmetric system, Theorem A shows that $\text{Con}(\mathbf{ZF} + \mathbf{VP})$ implies $\text{Con}(\mathbf{ZF} + \mathbf{VP} + \neg \mathbf{AC})$, while Theorem B shows that $\text{Con}(\mathbf{ZF} + \mathbf{VP})$ implies $\text{Con}(\mathbf{ZFC} + \mathbf{VP})$. Hence $\mathbf{AC}$ is independent of $\mathbf{ZF} + \mathbf{VP}$, relative to $\text{Con}(\mathbf{ZF} + \mathbf{VP})$.

Brooke-Taylor proved over $\mathbf{ZFC}$ that set forcing preserves $\mathbf{VP}$ for structures in a fixed finite language [5, Lemma 23 and Theorem 24]. Hayut and Karagila developed criteria for lifting elementary embeddings through symmetric extensions [9, Theorems 4.7 and 4.9]. Such a lift may exist in the generic extension while neither it nor its restriction to the two structures belongs to the symmetric extension. Our proof constructs in the symmetric extension the graph of that restriction.

Instead of choosing one name at each rank, the proof uses the set of all names of least possible rank. Applying $\mathbf{VP}$ to rank structures with these sets as distinguished constants gives an embedding that fixes the symmetric system. Proposition 2.4 then places the required restriction of its lift in the symmetric extension.

For Theorem B, we first obtain a model with unbounded $C^{(n)}$ -extendible cardinals for every standard n . Woodin's forcing at a sufficiently correct choiceless supercompact cardinal gives a $\mathbf{ZFC}$ rank model satisfying any prescribed finite subset of $\mathbf{ZFC} + \mathbf{VP}$. Compactness finishes the proof.

Sections 2 and 3 prove the lifting and preservation theorems. Section 4 derives a proper class of Löwenheim–Skolem cardinals, and Section 5 proves Theorem B. Sections 6–8 apply Theorem A

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to the Cohen, Feferman–Lévy, and full Solovay constructions. Section 9 asks whether Choice can instead be restored by class forcing over a given countable model while preserving VP.

All forcing notions have a largest condition $1_{\mathbb{P}}$. We write $\text{rank}_{\text{name}}(\tau)$ for the ground-model rank of a forcing name τ and f^*X for the direct image of X under f . We code structures so that the universe of a structure is contained in its transitive closure.

2. SYMMETRIC MODELS AND THE LIFTING LEMMA

Let $\mathcal{S} = \langle \mathbb{P}, \Gamma, \mathcal{F} \rangle$ be a symmetric system, where Γ is a group of automorphisms of $\mathbb{P}$ and $\mathcal{F}$ is a normal filter of subgroups of Γ . We write πp for the action of π on a condition p and $\pi \cdot \sigma$ for the induced recursive action on a name σ . The latter is defined by $\pi \cdot \sigma = \{ \langle \pi \cdot \tau, \pi p \rangle : \langle \tau, p \rangle \in \sigma \}$. Put $\text{sym}(\sigma) = \{ \pi \in \Gamma : \pi \cdot \sigma = \sigma \}$. A name is symmetric if its stabilizer belongs to $\mathcal{F}$. For a $\mathbb{P}$ -name σ , let $\text{cl}(\sigma)$ be the least set of names X with $\sigma \in X$ such that $v \in X$ whenever $\langle v, p \rangle \in \tau$ for some $\tau \in X$. A name σ is hereditarily symmetric if

$$(1) \quad \text{sym}(\tau) \in \mathcal{F} \quad \text{for every } \tau \in \text{cl}(\sigma).$$

Let HS denote the class of hereditarily symmetric names. Since $\sigma \in \text{cl}(\sigma)$, every hereditarily symmetric name is symmetric. We write $\Vdash_{\text{sym}}$ for the symmetric forcing relation. We use its definability, symmetry, and truth lemmas; see [12, Chapter 5], [13, Section 2], and [10, Lemma 2.2 and Theorem 2.5]. The action on names and the operation $\sigma \mapsto \text{cl}(\sigma)$ are uniformly first-order definable by well-founded recursion.

We shall repeatedly use the following restriction lemma. For a transitive set a , write $\text{Sat}(a)$ for the canonical satisfaction relation of $(a, \in)$.

Lemma 2.1. *Let $j : M \rightarrow N$ be elementary between transitive sets, and let $a \in M$ be transitive. If $\text{Sat}(a) \in M$, then $j \restriction a : (a, \in) \rightarrow (j(a), \in)$ is elementary.*

Proof. For each formula φ , relativize every quantifier to the parameter a . Transitivity of M makes the resulting formula compute truth in $(a, \in)$, and transitivity of N does the same for $j(a)$. Elementarity of j therefore gives, for every formula φ and every finite tuple $\bar{x}$ from a ,

$$(a, \in) \models \varphi(\bar{x}) \quad \text{if and only if} \quad (j(a), \in) \models \varphi(j(\bar{x})). \quad \square$$

The usual symmetric-model theorem holds over a ground model of ZF. The Power Set and Collection arguments are recorded because they require care without Choice.

Proposition 2.2. *Suppose that $V \models \text{ZF}$, that $\mathcal{S}$ is set-sized, and that $G \subseteq \mathbb{P}$ is V -generic. Then $W = \text{HS}^G = \{ \sigma^G : \sigma \in \text{HS} \}$ is a transitive model of ZF with $V \subseteq W \subseteq V[G]$.*

Proof. For $\sigma, \tau \in \text{HS}$, define the intersection name

$$\tau \cap^\bullet \sigma = \{ \langle \rho, r \rangle : \exists s (\langle \rho, s \rangle \in \sigma, r \leq s, \text{ and } r \Vdash_{\text{sym}} \rho \in \tau) \}.$$

Every $\pi \in \text{sym}(\sigma) \cap \text{sym}(\tau)$ fixes $\tau \cap^\bullet \sigma$, and every name appearing as the first coordinate of a pair in this name also appears as the first coordinate of a pair in σ . Hence $\tau \cap^\bullet \sigma \in \text{HS}$. Moreover, $\tau \cap^\bullet \sigma \subseteq \text{dom}(\sigma) \times \mathbb{P}$ and $(\tau \cap^\bullet \sigma)^G = \tau^G \cap \sigma^G$. Let $B_\sigma = \text{HS} \cap \mathcal{P}(\text{dom}(\sigma) \times \mathbb{P})$, which is a set by Separation, and define $\dot{Q}_\sigma = \{ \langle u, p \rangle : u \in B_\sigma \text{ and } p \Vdash_{\text{sym}} u \subseteq \sigma \}$. If $\pi \in \text{sym}(\sigma)$, then π permutes B_σ , because it preserves $\text{dom}(\sigma) \times \mathbb{P}$ and hereditary symmetry. For $u \in B_\sigma$ and $p \in \mathbb{P}$,

$$p \Vdash_{\text{sym}} u \subseteq \sigma \quad \text{if and only if} \quad \pi p \Vdash_{\text{sym}} \pi \cdot u \subseteq \sigma.$$

Thus $\text{sym}(\sigma) \leq \text{sym}(\dot{Q}_\sigma)$. Since $\text{sym}(\sigma) \in \mathcal{F}$ and every name occurring directly in $\dot{Q}_\sigma$ belongs to $B_\sigma \subseteq \text{HS}$, we have $\dot{Q}_\sigma \in \text{HS}$.

If $x \in \dot{Q}_\sigma^G$, then $x \in W$ and $x \subseteq \sigma^G$. Conversely, let $x \in W$ with $x \subseteq \sigma^G$, and choose $\tau \in \text{HS}$ with $\tau^G = x$. Then $\tau \cap^\bullet \sigma \in B_\sigma$ and $(\tau \cap^\bullet \sigma)^G = x$. Since $(\tau \cap^\bullet \sigma)^G \subseteq \sigma^G$, there is $p \in G$ such that $p \Vdash_{\text{sym}} \tau \cap^\bullet \sigma \subseteq \sigma$. Thus $x \in \dot{Q}_\sigma^G$, and consequently $\dot{Q}_\sigma^G = \mathcal{P}(\sigma^G) \cap W$. This proves Power Set in W .

To verify Collection, fix $\sigma, \vec{u} \in \text{HS}$ and suppose $W \models \forall x \in \sigma^G \exists y \varphi(x, y, \vec{u}^G)$. Choose $p \in G$ such that $p \Vdash_{\text{sym}} \forall x \in \sigma \exists y \varphi(x, y, \vec{u})$. Let $I = \{ \langle \rho, s, r \rangle : \langle \rho, s \rangle \in \sigma, r \leq p, \text{ and } r \leq s \}$. If $\langle \rho, s, r \rangle \in I$, then $r \Vdash_{\text{sym}} \rho \in \sigma$, so $r \Vdash_{\text{sym}} \exists y \varphi(\rho, y, \vec{u})$. There are therefore $t \leq r$ and $\tau \in \text{HS}$

such that $t \Vdash_{\text{sym}} \varphi(\rho, \tau, \vec{u})$. Collection and Separation in V give a set C of quintuples $\langle \rho, s, r, t, \tau \rangle$ such that every $\langle \rho, s, r \rangle \in I$ occurs as the first three coordinates of a member of C , with $t \leq r$ and $t \Vdash_{\text{sym}} \varphi(\rho, \tau, \vec{u})$.

Let $H = \text{sym}(\sigma) \cap \bigcap_i \text{sym}(u_i) \in \mathcal{F}$, put $b_0 = \{\langle \tau, t \rangle : \exists \rho, s, r \langle \rho, s, r, t, \tau \rangle \in C\}$, and set $b = \bigcup_{\pi \in H} \pi \cdot b_0$. Replacement shows that b is a set. Every $\xi \in H$ fixes b , since left multiplication by ξ permutes H . Thus $H \leq \text{sym}(b)$. Each name occurring directly in b has the form $\pi \cdot \tau$ with $\tau \in \text{HS}$, and normality of $\mathcal{F}$ gives $\pi \cdot \tau \in \text{HS}$. Since $H \in \mathcal{F}$, we have $b \in \text{HS}$. The identity belongs to H , so $b_0 \subseteq b$.

Let $x \in \sigma^G$. Choose $\langle \rho, s \rangle \in \sigma$ with $s \in G$ and $\rho^G = x$, and choose $r_0 \in G$ with $r_0 \leq p, s$. Define $D_{\rho, s} = \{t : \exists r, \tau \langle \rho, s, r, t, \tau \rangle \in C\}$. If $q \leq r_0$, then $\langle \rho, s, q \rangle \in I$, so C contains some $\langle \rho, s, q, t, \tau \rangle$ with $t \leq q$. Hence $D_{\rho, s}$ is dense below $r_0 \in G$, and genericity gives $t \in G \cap D_{\rho, s}$. Choose $\langle \rho, s, r, t, \tau \rangle \in C$. Then $\langle \tau, t \rangle \in b_0 \subseteq b$, so $\tau^G \in b^G$ and $W \models \varphi(x, \tau^G, \vec{u}^G)$.

Power Set and Collection are now established. For the remaining axioms, transitivity gives Extensionality and Foundation, canonical names give Empty Set, Pairing, Union, and Infinity, and symmetric subnames give Separation. Collection and Separation imply Replacement.

Thus $W \models \text{ZF}$. $\square$

Lemma 2.3. *Suppose that $V_\theta \models \text{ZF}$ and $\text{tc}(\{\mathcal{S}\}) \subseteq V_\theta$. If $\sigma \in V_\theta$ is a $\mathbb{P}$ -name, then hereditary symmetry computed in V_θ agrees with hereditary symmetry computed in V . Consequently, if $V_\eta \models \text{ZF}$ and $j : V_\theta \rightarrow V_\eta$ is elementary with $j(\mathcal{S}) = \mathcal{S}$, then $j(\sigma) \in \text{HS}$ whenever $\sigma \in \text{HS} \cap V_\theta$.*

Proof. Well-founded recursion is absolute to the transitive model V_θ , so the action on names and $\text{cl}(\tau)$ are computed correctly there. Thus $\text{cl}(\tau) \in V_\theta$. For every $\nu \in \text{cl}(\tau)$, V_θ and V compute $\text{sym}(\nu)$ from the same group Γ and agree whether it belongs to $\mathcal{F}$. Hence they agree on (1).

Now suppose that $j : V_\theta \rightarrow V_\eta$ is elementary, $j(\mathcal{S}) = \mathcal{S}$, and $\sigma \in \text{HS} \cap V_\theta$. By the first part, V_θ regards σ as hereditarily symmetric, so elementarity gives that V_η regards $j(\sigma)$ as hereditarily symmetric with respect to $\mathcal{S}$.

Since $j(\mathcal{S}) = \mathcal{S} \in V_\eta$ and V_η is transitive, $\text{tc}(\{\mathcal{S}\}) \subseteq V_\eta$. Applying the first part to V_η yields $j(\sigma) \in \text{HS}$ in V . $\square$

For names σ and τ , let $\langle \sigma, \tau \rangle^\bullet$ be the canonical name for their Kuratowski ordered pair. The construction is equivariant:

$$\pi \cdot \langle \sigma, \tau \rangle^\bullet = \langle \pi \cdot \sigma, \pi \cdot \tau \rangle^\bullet.$$

The canonical ordered-pair construction also preserves hereditary symmetry: if $\sigma, \tau \in \text{HS}$, every new name in its Kuratowski code is fixed by $\text{sym}(\sigma) \cap \text{sym}(\tau)$; hence $\langle \sigma, \tau \rangle^\bullet \in \text{HS}$.

Proposition 2.4. *Suppose that $V_\theta, V_\eta \models \text{ZF}$, and let $j : V_\theta \rightarrow V_\eta$ be elementary. Suppose that $j(z) = z$ for every $z \in \text{tc}(\{\mathcal{S}\})$. If $G \subseteq \mathbb{P}$ is V -generic and $W = \text{HS}^G$, then there is a function $E_j \in W$ such that*

$$E_j(\sigma^G) = j(\sigma)^G \quad (\sigma \in \text{HS} \cap V_\theta).$$

Thus, in $V[G]$, $E_j = j_G \upharpoonright \{\sigma^G : \sigma \in \text{HS} \cap V_\theta\}$, where $j_G : V_\theta[G] \rightarrow V_\eta[G]$ is the ordinary forcing lift of j , defined by $j_G(\sigma^G) = j(\sigma)^G$.

Proof. Since j fixes every member of $\text{tc}(\{\mathcal{S}\})$, it fixes $\mathbb{P}$ pointwise, so $j^*G = G$. Since G is V -generic, it is generic over both V_θ and V_η .

Define $j_G(\sigma^G) = j(\sigma)^G$ for every $\mathbb{P}$ -name $\sigma \in V_\theta$. The forcing theorem and elementarity show that this is well defined and that

$$p \Vdash_{V_\theta} \varphi(\vec{\sigma}) \quad \text{if and only if} \quad p \Vdash_{V_\eta} \varphi(j(\vec{\sigma}))$$

for every formula φ , condition $p \in \mathbb{P}$, and finite tuple $\vec{\sigma}$ of names from V_θ . Thus $j_G : V_\theta[G] \rightarrow V_\eta[G]$ is elementary. Put $D_\theta = \text{HS} \cap V_\theta$. Hereditary symmetry is first-order definable from $\mathcal{S}$, so D_θ is a set by Separation. Lemma 2.3 gives $j(\sigma) \in \text{HS}$ for every $\sigma \in D_\theta$.

The action of Γ on $\mathbb{P}$ -names is defined by well-founded recursion. Hence, for $\pi \in \Gamma$ and every $\mathbb{P}$ -name $\sigma \in V_\theta$, elementarity and $j(\pi) = \pi$ give

$$(2) \quad j(\pi \cdot \sigma) = j(\pi) \cdot j(\sigma) = \pi \cdot j(\sigma).$$

Moreover, $\text{cl}(\pi \cdot \sigma) = \pi \text{cl}(\sigma)$, and $\text{sym}(\pi \cdot \nu) = \pi \text{sym}(\nu) \pi^{-1}$. Since $\mathcal{F}$ is normal, σ is hereditarily symmetric if and only if $\pi \cdot \sigma$ is hereditarily symmetric. As π bijects the $\mathbb{P}$ -names in V_θ , it follows that $\pi \text{``} D_\theta = D_\theta$.

The required graph name is

$$\dot{E}_j = \{ \langle \langle \sigma, j(\sigma) \rangle^\bullet, 1_\mathbb{P} \rangle : \sigma \in D_\theta \}.$$

Since D_θ is a set, $\dot{E}_j$ is a $\mathbb{P}$ -name, and every pair name occurring in it belongs to HS. For each $\pi \in \Gamma$, equivariance of the ordered-pair name, $\pi 1_\mathbb{P} = 1_\mathbb{P}$, equation (2), and $\pi \text{``} D_\theta = D_\theta$ give

$$\pi \cdot \dot{E}_j = \{ \langle \langle \pi \cdot \sigma, j(\pi \cdot \sigma) \rangle^\bullet, 1_\mathbb{P} \rangle : \sigma \in D_\theta \} = \dot{E}_j.$$

Thus $\text{sym}(\dot{E}_j) = \Gamma \in \mathcal{F}$, so $\dot{E}_j \in \text{HS}$.

Let $E_j = \dot{E}_j^G$. Then

$$E_j = \{ \langle \sigma^G, j(\sigma)^G \rangle : \sigma \in D_\theta \}.$$

If $\sigma^G = \tau^G$, the well-definedness of j_G gives $j(\sigma)^G = j(\tau)^G$. Hence E_j is a function in W , with

$$E_j(\sigma^G) = j(\sigma)^G \quad (\sigma \in D_\theta). \quad \square$$

3. PRESERVATION

Lemma 3.1. *Let $\theta > \omega$ be a limit ordinal. If no function $f : a \rightarrow \theta$ with $a \in V_\theta$ has cofinal range, then $V_\theta \models \text{ZF}$. The hypothesis on θ is Π_1 .*

Proof. Only Replacement and Collection require proof. Suppose $a, p \in V_\theta$ and $V_\theta \models \forall x \in a \exists! y \varphi(x, y, p)$. Replacement in V gives the corresponding function F . If the ranks of its values were unbounded in θ , then $x \mapsto \text{rank}(F(x))$ would be cofinal in θ , contrary to the hypothesis. Hence $F \text{``} a \in V_\theta$.

The same rank argument proves Collection. Suppose $V_\theta \models \forall x \in a \exists y \varphi(x, y, p)$. For each $x \in a$, take the least rank of a suitable $y \in V_\theta$. These ranks are bounded below θ . If $\beta < \theta$ lies above them, Separation applied to V_β gives a set containing a suitable y for every $x \in a$. Finally, $x \in V_\theta$ is Δ_1 in θ , and “ $f : a \rightarrow \theta$ is cofinal” is Δ_0 in f, a, θ . Thus the hypothesis is Π_1 . $\square$

Lemma 3.2. *Assume $\text{ZF} + \text{VP}(\Pi_1)$. Above every ordinal γ there is a limit ordinal κ such that no map from any $a \in V_\kappa$ is cofinal in κ . Consequently $V_\kappa \models \text{ZF}$. Under AC, such a κ may be chosen measurable.*

Proof. Fix γ and put $\bar{\gamma} = \max\{\gamma, \omega\}$. For $\lambda > \bar{\gamma}$, consider

$$\mathcal{B}_\lambda = \langle V_{\lambda+\omega}, \in, \lambda, (z)_{z \in V_{\bar{\gamma}+1}} \rangle$$

in the set-sized language with membership, one distinguished constant, and constants for $V_{\bar{\gamma}+1}$. The structures $\mathcal{B}_\lambda$ form a Π_1 -definable proper class, using $V_{\bar{\gamma}+1}$ as a parameter: the hierarchy equation for the domain has a Π_1 definition, ordinal addition is Δ_1 , and the distinguished ordinal and interpretation tables are extracted by bounded quantifiers from the structure code. Applying $\text{VP}(\Pi_1)$ to this class gives distinct ordinals λ, μ and an elementary embedding $j : \mathcal{B}_\lambda \rightarrow \mathcal{B}_\mu$. By elementarity and the choice of constants, $j(\lambda) = \mu$, while $j(z) = z$ for every $z \in V_{\bar{\gamma}+1}$. Since $\lambda \neq \mu$, the embedding moves an ordinal. Let κ be the least ordinal moved by j . Then $\bar{\gamma} < \kappa \leq \lambda$, $j(\kappa) > \kappa$, and a rank induction shows that $j(x) = x$ for every $x \in V_\kappa$.

Suppose that $f : a \rightarrow \kappa$ is cofinal for some $a \in V_\kappa$. The embedding fixes a and each of its elements, so $j(f) : a \rightarrow j(\kappa)$ is cofinal and $j(f)(x) = f(x) < \kappa$ for every $x \in a$. This is impossible because $j(\kappa) > \kappa$. Thus no map from a member of V_κ is cofinal in κ . The ordinal κ cannot be a successor, since j fixes every smaller ordinal. As $\kappa > \bar{\gamma} \geq \omega$, Lemma 3.1 applies, so $V_\kappa \models \text{ZF}$.

Assume now that AC holds and define $U = \{X \subseteq \kappa : \kappa \in j(X)\}$.

Elementarity shows that U is an ultrafilter on κ . Because j fixes every ordinal below κ , U is nonprincipal. Moreover, U is κ -complete, since $V_{\lambda+\omega}$ contains every sequence of fewer than κ subsets of κ . Under Choice, κ must be a cardinal. If it were not, then κ could be partitioned into fewer than κ singletons, contradicting the κ -completeness of the nonprincipal ultrafilter U . Thus U is a nonprincipal κ -complete ultrafilter on the cardinal κ , and therefore κ is measurable. $\square$

Theorem 3.3. *Let $V \models \text{ZF} + \text{VP}$, with VP understood in the arbitrary-language sense fixed in the Introduction. Let $\mathcal{S} = \langle \mathbb{P}, \Gamma, \mathcal{F} \rangle \in V$ be a set-sized symmetric system, and let $G \subseteq \mathbb{P}$ be V -generic. Then*

$$W = \text{HS}^G \models \text{ZF} + \text{VP}.$$

Proof. Proposition 2.2 gives $W \models \text{ZF}$. Fix a formula $\varphi(x, y)$, a parameter $a \in W$, and a set-sized language $L \in W$ such that $\mathcal{A} = \{M \in W : W \models \varphi(M, a)\}$ is a proper class of L -structures. Choose $\dot{a}, \dot{L} \in \text{HS}$ with $\dot{a}^G = a$ and $\dot{L}^G = L$. For $p \in \mathbb{P}$, let R_p be the class of ordinals α for which there is a name $\tau \in \text{HS}$ satisfying

$$(3) \quad p \Vdash_{\text{sym}} \text{"}\tau \text{ is an } \dot{L}\text{-structure, } \varphi(\tau, \dot{a}), \text{ and } \text{rank}(\tau) = \check{\alpha} \text{"}$$

In (3), $\text{rank}(\tau)$ denotes the rank of the value of τ in the generic extension. By contrast, $\text{rank}_{\text{name}}(\tau)$ is the von Neumann rank of the name τ itself.

There is a condition $q \in G$ for which R_q is unbounded in Ord. Suppose not. Let $D_{\text{bd}} = \{p \in \mathbb{P} : R_p \text{ is bounded}\}$ and, for $p \in D_{\text{bd}}$, let $b(p)$ be the least ordinal such that $R_p \subseteq b(p)$. The domain D_{bd} is a subset of the set $\mathbb{P}$, and definability of symmetric forcing together with Replacement makes b a set function in V . Our supposition gives $G \subseteq D_{\text{bd}}$.

In $V[G]$, set $\delta = \sup b''G$. This is an ordinal, so preservation of ordinals gives $\delta \in V \subseteq W$. If $M \in \mathcal{A}$, choose $\tau \in \text{HS}$ with $\tau^G = M$ and let $\alpha = \text{rank}(M)$. By the symmetric truth lemma, some $p \in G$ forces the assertion in (3); hence $\alpha < b(p) \leq \delta$. Thus $\mathcal{A} \subseteq V_\delta^W$, and Separation in W makes $\mathcal{A}$ a set, a contradiction.

For $\alpha \in R_q$, let $\Psi_\alpha(\tau)$ assert that $\tau \in \text{HS}$ and that (3) holds with q in place of p . We use the following definitions:

$$\nu_\alpha = \min\{\text{rank}_{\text{name}}(\tau) : \Psi_\alpha(\tau)\}, \quad X_\alpha = \{\tau \in V_{\nu_\alpha+1} : \text{rank}_{\text{name}}(\tau) = \nu_\alpha \text{ and } \Psi_\alpha(\tau)\}.$$

The ordinal ν_α exists, X_α is a nonempty set by Separation, and both are uniformly definable from α and the fixed parameters.

Choose $\rho > \omega$ such that $\text{tc}(\{\mathcal{S}, \dot{a}, \dot{L}\}) \subseteq V_\rho$. By Lemma 3.2, for each $\alpha \in R_q$ there are arbitrarily high θ with $V_\theta \models \text{ZF}$. The relevant rank is

$$\theta_\alpha = \min\{\theta : V_\theta \models \text{ZF} \text{ and } \theta > \max\{\rho, \alpha, \text{rank}(X_\alpha)\} + \omega\}.$$

In the common language consisting of a binary relation symbol, two distinguished constant symbols, and constant symbols c_z for $z \in V_\rho$, use the structure

$$\mathcal{M}_\alpha = \langle V_{\theta_\alpha}, \in, \alpha, X_\alpha, (c_z)_{z \in V_\rho} \rangle \quad (\alpha \in R_q).$$

The structures $\mathcal{M}_\alpha$ form a definable proper class since R_q is unbounded. By VP, there is an elementary embedding $j : \mathcal{M}_\alpha \rightarrow \mathcal{M}_\beta$ with $\alpha \neq \beta$. The distinguished constants give $j(\alpha) = \beta$, $j(X_\alpha) = X_\beta$, and $j(z) = z$ for every $z \in V_\rho$.

Fix $\tau \in X_\alpha$. Then $j(\tau) \in X_\beta$, and the definitions of X_α and X_β give

$$(4) \quad q \Vdash_{\text{sym}} \text{"}\tau \text{ and } j(\tau) \text{ are } \dot{L}\text{-structures satisfying } \varphi(-, \dot{a}) \text{ with ranks } \check{\alpha} \text{ and } \check{\beta} \text{"}$$

Since j fixes V_ρ pointwise, it fixes every member of $\text{tc}(\{\mathcal{S}, \dot{a}, \dot{L}\})$.

Apply Proposition 2.4 to obtain the elementary lift $j_G : V_{\theta_\alpha}[G] \rightarrow V_{\theta_\beta}[G]$ and a function $E_j \in W$ agreeing with j_G on values of names in $\text{HS} \cap V_{\theta_\alpha}$. Put $M = \tau^G$ and $N = j(\tau)^G$. Then $j_G(M) = N$.

If $s \in L = \dot{L}^G$, then $s = \sigma^G$ for some $\langle \sigma, p \rangle \in \dot{L}$ with $p \in G$. Since $\sigma \in \text{cl}(\dot{L}) \subseteq V_\rho$, we have $j(\sigma) = \sigma$, and hence $j_G(s) = s$. Thus j_G fixes L pointwise. Moreover, $j(\dot{L}) = \dot{L}$ and $j_G(L) = L$.

By the coding convention of the Introduction, $|M| \subseteq \text{tc}(M)$. An induction on name evaluation shows that every element of $\text{tc}(M)$ is σ^G for some $\sigma \in \text{cl}(\tau)$. Since $\tau \in \text{HS} \cap V_{\theta_\alpha}$, every such σ belongs to $\text{HS} \cap V_{\theta_\alpha}$. Hence $|M| \subseteq \text{dom}(E_j)$, and $e = E_j \restriction |M|$ belongs to W . In $V[G]$, the map e is $j_G \restriction |M|$. Since the universe of a structure is uniformly definable from its fixed set code,

$$j_G(|M|) = |j_G(M)| = |N|, \quad j_G''|M| \subseteq |N|.$$

Satisfaction for set-sized structures is absolute between the transitive models under consideration. Since j_G fixes L pointwise, elementarity gives

$$M \models \psi(\bar{m}) \quad \text{if and only if} \quad N \models \psi(e(\bar{m}))$$

for every L -formula ψ and every finite tuple $\bar{m}$ from $|M|$. Thus W regards e as an L -elementary embedding from M to N .

Finally, $q \in G$ and (4) imply that $M, N \in \mathcal{A}$, with ranks α and β . Thus $M \neq N$. This proves the arbitrary instance of VP, so $W \models \text{ZF} + \text{VP}$. $\square$

Corollary 3.4. *Let $n \geq 1$ be standard. If $V \models \text{ZF} + \text{VP}(\Pi_n)$, $\mathcal{S} = \langle \mathbb{P}, \Gamma, \mathcal{F} \rangle \in V$ is a set-sized symmetric system, $G \subseteq \mathbb{P}$ is V -generic, and $W = \text{HS}^G$, then*

$$W \models \text{ZF} + \text{VP}(\Pi_n).$$

Proof. For each fixed standard formula of positive Lévy level $k \geq 1$, symmetric forcing has a definition of the same polarity and level, uniformly in the ground model and the symmetric system. The bounded base has a Δ_1 definition. To obtain it, use set certificates for the hereditary name closure, the action and stabilizer tables, and the atomic forcing recursion. Existence and uniqueness of these recursion tables give the two Σ_1 and Π_1 presentations; bounded quantifiers range over subnames and conditions. At an unbounded existential step the clause is that below every condition there is a stronger condition and an HS name forcing the matrix. The condition quantifiers are bounded by the set $\mathbb{P}$, and the name quantifier gives the required Σ_k presentation. Negation gives the dual Π_k presentation. Collection supplies the set certificates used in this induction; no selection of one name for each condition is required.

We use the following elementary coding observation. If $S(z)$ is Σ_k , $k \geq 1$, write it as $\exists u \psi(z, u)$ with ψ in Π_{k-1} . When $S(z)$ holds, let ξ be the least rank of such a u and let C be the set of all such u of rank ξ . There is a uniform Π_k formula $\chi_S(z, c)$ saying that

$$c = \langle \xi, V_\xi, V_{\xi+1}, C \rangle$$

has these properties. Indeed, once the two rank segments are included in c , all quantifiers used for minimality and for the exact description of C are bounded; the equations defining the rank segments are Π_1 . Hence c is unique and $\chi_S(z, c)$ implies $S(z)$.

The Π_n argument uses this observation twice: to record the least name rank together with all witnesses at that rank, and to record the least rank satisfying Lemma 3.1.

Lemma 3.2 applies because $\text{VP}(\Pi_n)$ includes $\text{VP}(\Pi_1)$. It gives unboundedly many ordinals satisfying the Π_1 condition of Lemma 3.1. Write $B(W, \theta, b)$ for the bounded assertion that, for every $b < \eta < \theta$, either η is not a limit above ω , or W contains a cofinal map $f : a \rightarrow \eta$ and a transitive rank-table certificate showing $\text{rank}(a) < \eta$. If θ is the least rank-criterion height above b , then $B(V_\theta, \theta, b)$ holds: for each $\eta < \theta$, the map and a rank table on V_η belong to V_θ , since θ is a limit. Conversely, any witness to B certifies failure of the criterion at every earlier candidate. Apply the preceding least-witness coding observation to $\exists W B(W, \theta, b)$. The resulting unique certificate d , together with the positive criterion at θ , gives a Π_1 definition of (θ, d) from b . The available witness V_θ has rank θ , so the least witness rank is at most θ and the entire certificate d belongs to $V_{\theta+\omega}$. The satisfaction relation of V_θ also belongs to that rank.

Now let a Π_n formula $\varphi(x, a)$ define a proper class $\mathcal{A}$ of structures in a set-sized language L in W . Choose names $\dot{a}, \dot{L} \in \text{HS}$. Define R_p as in the proof of Theorem 3.3. The same boundedness argument gives $q \in G$ for which R_q is unbounded. Let $\Psi(\alpha, \tau)$ say that $\tau \in \text{HS}$ and that q forces τ to be an $\dot{L}$ -structure of rank α satisfying $\varphi(\tau, \dot{a})$. By the first paragraph, Ψ is Π_n .

For $\alpha \in R_q$, let ν_α be the least von Neumann rank of a name τ satisfying $\Psi(\alpha, \tau)$. We use the following notation:

$$U_\alpha = V_{\nu_\alpha}, \quad U_\alpha^+ = V_{\nu_\alpha+1}, \quad X_\alpha = \{\tau \in U_\alpha^+ : \text{rank}_{\text{name}}(\tau) = \nu_\alpha \text{ and } \Psi(\alpha, \tau)\}.$$

The assertions

$$\forall \tau \in U_\alpha \neg \Psi(\alpha, \tau) \quad \text{and} \quad \forall \tau \in U_\alpha^+ ((\text{rank}_{\text{name}}(\tau) = \nu_\alpha \wedge \Psi(\alpha, \tau)) \longrightarrow \tau \in X_\alpha)$$

are Σ_n . Apply the preceding coding observation separately to them, and let c_α be the ordered pair of the resulting codes. Its first component records that no name of rank below ν_α satisfies

$\Psi(\alpha, \cdot)$; its second records that every name of rank ν_α satisfying $\Psi(\alpha, \cdot)$ belongs to X_α . Keeping c_α as a distinguished component lets the defining formula state both coded Π_n conditions without another existential quantifier. Together with the rank-segment equations and the assertion that X_α is the nonempty set of rank- ν_α names satisfying $\Psi(\alpha, \cdot)$, these clauses give a Π_n definition from α of

$$\langle \nu_\alpha, U_\alpha, U_\alpha^+, X_\alpha, c_\alpha \rangle.$$

Choose $\rho > \omega$ with $\text{tc}(\{\mathcal{S}, \dot{a}, \dot{L}\}) \subseteq V_\rho$. Let θ_α be the least limit ordinal above $\max\{\rho, \alpha, \text{rank}(\langle \nu_\alpha, U_\alpha, U_\alpha^+, X_\alpha, c_\alpha \rangle) + \omega$ admitting no cofinal map $f : a \rightarrow \theta_\alpha$ with $a \in V_{\theta_\alpha}$, and let d_α be the unique code witnessing this leastness in the Π_1 definition described above. In the set-sized language with membership, finitely many distinguished constants, and constants for V_ρ , let

$$\mathcal{N}_\alpha = \langle V_{\theta_\alpha + \omega}, \in, \alpha, \nu_\alpha, U_\alpha, U_\alpha^+, X_\alpha, c_\alpha, \theta_\alpha, d_\alpha, (z)_{z \in V_\rho} \rangle.$$

The rank-segment equations and the definition of θ_α are Π_1 , the least-name-rank clause is Π_n , and the other clauses are bounded or Δ_1 . Thus $\{\mathcal{N}_\alpha : \alpha \in R_q\}$ is a Π_n -definable proper class: R_q is unbounded, and the tuple is unique for each α .

Apply $\text{VP}(\Pi_n)$ in V . There are distinct $\mathcal{N}_\alpha, \mathcal{N}_\beta$ and an elementary embedding $j : \mathcal{N}_\alpha \rightarrow \mathcal{N}_\beta$. Uniqueness gives $\alpha \neq \beta$, and the distinguished constants give $j(\alpha) = \beta$, $j(X_\alpha) = X_\beta$, and $j(\theta_\alpha) = \theta_\beta$. Since $\text{Sat}(V_{\theta_\alpha}) \in V_{\theta_\alpha + \omega}$, Lemma 2.1 shows that $j_0 = j \upharpoonright V_{\theta_\alpha} : V_{\theta_\alpha} \rightarrow V_{\theta_\beta}$ is elementary. Both ranks satisfy ZF, and the constants from V_ρ make j_0 fix the symmetric system, every condition, and the names $\dot{a}$ and $\dot{L}$.

Choose $\tau \in X_\alpha$. Then $j_0(\tau) \in X_\beta$, and q forces that the two names denote members of $\mathcal{A}$ of ranks α and β . Proposition 2.4 lifts j_0 through G . The restriction of the lifted map to the universe of τ^G lies in W . The lift fixes L pointwise, so this restriction is an elementary L -embedding from τ^G to $j_0(\tau)^G$. Since $q \in G$ and $\alpha \neq \beta$, these are distinct members of $\mathcal{A}$. $\square$

This corollary, together with Theorem 3.3, proves Theorem A.

4. LÖWENHEIM–SKOLEM CARDINALS

We use Usuba's definitions of weakly LS and LS cardinals [21, Definition 1.1].

Definition 4.1. Let κ be an uncountable cardinal.

- (1) κ is *weakly Löwenheim–Skolem*, or *weakly LS*, if for every $\gamma < \kappa$, every $\alpha \geq \kappa$, and every $x \in V_\alpha$, there is $X \prec V_\alpha$ such that $V_\gamma \subseteq X$, $x \in X$, and the transitive collapse of X belongs to V_κ .
- (2) κ is *Löwenheim–Skolem*, or *LS*, if for every $\gamma < \kappa$, every $\alpha \geq \kappa$, and every $x \in V_\alpha$, there are $\beta \geq \alpha$ and $X \prec V_\beta$ such that $V_\gamma \subseteq X$, $x \in X$, the transitive collapse of X belongs to V_κ , and

$$V_\gamma(X \cap V_\alpha) \subseteq X.$$

The last clause means that every function from V_γ into $X \cap V_\alpha$ belongs to X .

Lemma 4.2. A limit of LS cardinals is LS, and a limit of weakly LS cardinals is weakly LS.

Proof. Suppose that κ is a limit of LS cardinals. It is uncountable, and it is a cardinal: if $b : \kappa \rightarrow \alpha$ were a bijection for some $\alpha < \kappa$, an LS cardinal μ with $\alpha < \mu < \kappa$ would make $b \upharpoonright \mu$ an injection from the initial ordinal μ into the smaller ordinal α , a contradiction. Now fix $\gamma < \kappa \leq \alpha$ and $x \in V_\alpha$. Choose an LS cardinal μ with $\gamma < \mu < \kappa$, and apply its LS property to (γ, α, x) . The submodel supplied by the LS property of μ has transitive collapse in $V_\mu \subseteq V_\kappa$, so it also witnesses the LS property for κ . The weakly LS argument is the same. $\square$

For $n \geq 1$, put $C^{(n)} = \{\alpha : V_\alpha \prec_{\Sigma_n} V\}$.

Theorem 4.3. The theory $\text{ZF} + \text{VP}$ proves that there is a proper class of LS cardinals.

Proof. VP includes $\text{VP}(\Pi_2)$. By [16, Theorem 3.1], there is a proper class of 1-choiceless extendible ordinals. Each is a cardinal by the observation following [16, Definition 2.1], and [16, Theorem 2.8] identifies them with the 2-choiceless supercompact cardinals. It is therefore enough to prove that every 2-choiceless supercompact cardinal is LS.

Let κ be 2-choiceless supercompact, and fix $\delta < \kappa \leq \alpha$ and $x \in V_\alpha$. Choose $\lambda \in C^{(2)}$ above $\alpha + \omega$. Apply [16, Definition 2.3] with lower bound δ and parameter $a = \langle \alpha, x, \alpha + \omega \rangle$. We obtain $\bar{\lambda} < \kappa$ in $C^{(2)}$, an element $\bar{a} \in V_{\bar{\lambda}}$, and an elementary embedding $j : V_{\bar{\lambda}} \rightarrow V_\lambda$ such that $\text{crit}(j) > \delta$ and $j(\bar{a}) = a$. Decoding $\bar{a}$ gives ordinals $\bar{\alpha} < \bar{\lambda}$ and $\bar{x} \in V_{\bar{\alpha}}$ with $\bar{a} = \langle \bar{\alpha}, \bar{x}, \bar{\alpha} + \omega \rangle$, $j(\bar{\alpha}) = \alpha$, $j(\bar{x}) = x$, and $\bar{\alpha} + \omega < \bar{\lambda}$. Moreover $\bar{\alpha} > \delta$, because j fixes every ordinal below $\text{crit}(j)$ whereas $j(\bar{\alpha}) = \alpha \geq \kappa > \delta$.

Put $X = j''V_{\bar{\lambda}}$. Then $X \prec V_\lambda$, $V_\delta \subseteq X$, $x \in X$, and the transitive collapse of X is $V_{\bar{\lambda}} \in V_\kappa$. Moreover, $X \cap V_\alpha = j''V_{\bar{\alpha}}$. If $f : V_\delta \rightarrow X \cap V_\alpha$, let $\bar{f}(u)$ be the unique $y \in V_{\bar{\alpha}}$ such that $j(y) = f(u)$. Replacement gives $\bar{f}$, and $\bar{\alpha} + \omega < \bar{\lambda}$ implies $\bar{f} \in V_{\bar{\lambda}}$. Since j fixes V_δ pointwise, $j(\bar{f}) = f$. Thus $V_\delta(X \cap V_\alpha) \subseteq X$, and κ is LS. $\square$

Corollary 4.4. *Above every ordinal there is a singular LS cardinal of cofinality ω .*

Proof. Starting above the given ordinal, recursively take the least LS cardinal above the preceding one and let κ be the supremum of the resulting strictly increasing ω -sequence. Lemma 4.2 makes κ LS, and the sequence witnesses $\text{cf}(\kappa) = \omega$. $\square$

Corollary 4.5. *The theory $\text{ZF} + \text{VP}$ proves that there is a definable class forcing which forces ZFC.*

Proof. Combine Theorem 4.3 with Usuba's Choice-restoration theorem [21, Corollary 4.8]; the resulting class forcing forces ZFC. $\square$

5. CONSISTENCY TRANSFER TO CHOICE

Proposition 5.4 reduces the forward consistency implication to a model with unbounded $C^{(n)}$ -extendible cardinals for every standard n . If this model satisfies Choice, it already satisfies $\text{ZFC} + \text{VP}$. Otherwise, for each finite $F \subseteq \text{ZFC} + \text{VP}$, Theorem 5.11 produces a ZFC rank model satisfying F . Compactness then yields a model of $\text{ZFC} + \text{VP}$.

Choiceless extendibility. Following [16, Definition 6.6], let $E_n(\kappa)$ assert that κ is a cardinal and that, whenever $\lambda > \kappa$ belongs to $C^{(n)}$, there are $\theta > \lambda$ in $C^{(n)}$ and an elementary embedding $j : V_\lambda \rightarrow V_\theta$ with $\text{crit}(j) = \kappa$ and $j(\kappa) > \lambda$. We call such a κ $C^{(n)}$ -extendible; over ZFC, this agrees with Bagaria's notion. Let UE, for unbounded extendibility, be the recursive scheme asserting that the E_n -cardinals are unbounded for every $n \geq 1$. This fixed-critical-point notion differs from the variable-critical-point supercompactness used in Theorem 4.3.

Lemma 5.1. *Over ZF, UE implies VP.*

Proof. Let $\mathcal{A} = \{M : \varphi(M, a)\}$ be a proper class of structures in a set-sized language L , and put $\rho = \text{rank}(\text{tc}(\{a, L\}))$. Choose n large enough that $C^{(n)}$ -correct ranks are correct about φ , the coding of L -structures, and satisfaction for set-sized L -structures. Choose a $C^{(n)}$ -extendible cardinal $\kappa > \rho$ and $M \in \mathcal{A}$ with $\alpha = \text{rank}(M) > \kappa$. Let $\lambda \in C^{(n)}$ lie above $\alpha + \omega$, and take $j : V_\lambda \rightarrow V_\theta$ witnessing $C^{(n)}$ -extendibility of κ , with $\theta \in C^{(n)}$. The parameters a and L are fixed. Correctness of the source and target ranks, followed by elementarity, gives $j(M) \in \mathcal{A}$, and $j \upharpoonright |M| : M \rightarrow j(M)$ is L -elementary. Finally,

$$\text{rank}(j(M)) = j(\alpha) \geq j(\kappa) > \lambda > \alpha,$$

so the two structures are distinct. $\square$

We recall that a nonzero cardinal δ is rank-Berkeley if, for every $\zeta < \delta$ and every $\Theta > \delta$, there is a nontrivial elementary self-embedding $j : V_\Theta \rightarrow V_\Theta$ with $\zeta < \text{crit}(j) < \delta$ and $j(\delta) = \delta$ [16, Definition 4.4].

Lemma 5.2. *ZF proves that the existence of a rank-Berkeley cardinal implies the existence of a set model of*

$$\text{ZF} + \text{UE} + \text{VP}.$$

Proof. Let δ be rank-Berkeley. Taking $\zeta = 0 < \delta$, choose $\Theta > \delta + \omega$ and a nontrivial elementary embedding $j : V_\Theta \rightarrow V_\Theta$ with critical point $\kappa_0 < \delta$ and $j(\delta) = \delta$. Put $\kappa_{m+1} = j(\kappa_m)$ and $\lambda = \sup_{m < \omega} \kappa_m$. Inductively, $\kappa_m < \delta$ for all m , so $\lambda \leq \delta < \Theta$. The sequence $s = \langle \kappa_m : m < \omega \rangle$ belongs to V_Θ , and $j(s)$ is its tail. Since $\lambda = \sup \text{ran}(s)$ is a first-order relation between objects in the domain,

$$j(\lambda) = \sup \text{ran}(j(s)) = \sup_{m < \omega} \kappa_{m+1} = \lambda.$$

Hence $j \upharpoonright V_\lambda : V_\lambda \rightarrow V_\lambda$ is elementary.

If $f : a \rightarrow \kappa_0$ were cofinal for some $a \in V_{\kappa_0}$, then $j(f)$ would be cofinal in κ_1 although its range remained below κ_0 . Lemma 3.1 therefore gives $V_{\kappa_0} \models \text{ZF}$. Since $j \upharpoonright V_{\kappa_0}$ is the identity, Lemma 2.1 gives $V_{\kappa_0} \prec V_{\kappa_1}$. Applying j^m to this assertion gives

$$V_{\kappa_m} \prec V_{\kappa_{m+1}} \quad (m < \omega).$$

The elementary-chain theorem yields $V_{\kappa_m} \prec V_\lambda$ for every m , and $V_\lambda \models \text{ZF}$.

Fix $n \geq 1$ and $i < \omega$. We show inside V_λ that κ_i is $C^{(n)}$ -extendible. For $\mu < \lambda$, internal membership in $C^{(n)}$ is characterized by

$$\mu \in (C^{(n)})^{V_\lambda} \quad \text{if and only if} \quad V_\mu \prec_{\Sigma_n} V_{\kappa_r} \text{ for some } r \text{ with } \mu < \kappa_r,$$

because every V_{κ_r} is elementary in V_λ . Let $\mu > \kappa_i$ belong to $(C^{(n)})^{V_\lambda}$, and choose $r \geq i$ with $\mu + \omega < \kappa_r$ and $V_\mu \prec_{\Sigma_n} V_{\kappa_r}$. Choose $s > r$ and $t > s$. The set-sized restriction $e = j^t \upharpoonright V_{\kappa_s} : V_{\kappa_s} \rightarrow V_{\kappa_{s+t}}$ is elementary by Lemma 2.1, has rank below $\kappa_{s+t} + \omega$, and therefore belongs to V_λ . Put $k = j^i(e)$, where this denotes the image of the set e under j^i . Then $k \in V_\lambda$, $k : V_{\kappa_{s+i}} \rightarrow V_{\kappa_{s+t+i}}$, $\text{crit}(k) = \kappa_i$, and $k(\kappa_i) = \kappa_{t+i} > \mu$. Since $r \geq i$, we have $\kappa_r = j^i(\kappa_{r-i})$, and hence

$$k(\kappa_r) = j^i(e(\kappa_{r-i})) = \kappa_{r+t}.$$

Applying k to $V_\mu \prec_{\Sigma_n} V_{\kappa_r}$ gives

$$V_{k(\mu)} \prec_{\Sigma_n} V_{\kappa_{r+t}}.$$

As $V_{\kappa_{r+t}} \prec V_\lambda$, the displayed characterization shows that $k(\mu) \in (C^{(n)})^{V_\lambda}$. Because k has critical point κ_i , V_λ regards κ_i as a cardinal. Indeed, a surjection $f : \xi \rightarrow \kappa_i$ with $\xi < \kappa_i$ would be carried to a surjection from ξ onto $k(\kappa_i)$, although $k(f)(\zeta) = f(\zeta) < \kappa_i$ for every $\zeta < \xi$. The map $k \upharpoonright V_\mu \in V_\lambda$ is therefore the embedding required by $E_n(\kappa_i)$ for the chosen μ . Thus every κ_i is $C^{(n)}$ -extendible in V_λ . Since the critical sequence is cofinal in λ and n was arbitrary, $V_\lambda \models \text{UE}$. Lemma 5.1 now gives $V_\lambda \models \text{VP}$. $\square$

Lemma 5.3. *If $m > n \geq 1$, then every $C^{(m)}$ -extendible cardinal is $C^{(n)}$ -extendible.*

Proof. It is enough to consider $m = n + 1$. Let κ be $C^{(n+1)}$ -extendible, and let $\lambda > \kappa$ belong to $C^{(n)}$. Choose $\mu > \lambda$ in $C^{(n+1)}$ and an elementary embedding $j : V_\mu \rightarrow V_\nu$ with $\text{crit}(j) = \kappa$, $j(\kappa) > \mu$, and $\nu \in C^{(n+1)}$. Because $\mu \in C^{(n+1)}$, we have $V_\mu \models \lambda \in C^{(n)}$. Elementarity therefore gives

$$V_\nu \models j(\lambda) \in C^{(n)}.$$

Since $\nu \in C^{(n+1)}$, $j(\lambda) \in C^{(n)}$ also holds in V . For $n = 1$, this follows from the Δ_2 definition of $C^{(1)}$, while for $n \geq 2$ it follows from the usual Π_n definition of $C^{(n)}$. Thus $j(\lambda) \in C^{(n)}$, and $j \upharpoonright V_\lambda : V_\lambda \rightarrow V_{j(\lambda)}$ witnesses the required $C^{(n)}$ -extendibility of κ . $\square$

Proposition 5.4.

$\text{Con}(\text{ZF} + \text{VP}) \text{ implies } \text{Con}(\text{ZF} + \text{VP} + \text{UE} + \text{"there are no rank-Berkeley cardinals"})$.

Proof. Put $T = \text{ZF} + \text{VP}$. This is a recursively axiomatized theory: the instances of the parameterized Vopěnka scheme are generated effectively from formula codes. The assertion RB that a rank-Berkeley cardinal exists is a single first-order sentence, since elementarity between set ranks is expressed by their canonical satisfaction relations. The proof of Lemma 5.2 can be formalized in T , uniformly in formula codes. Hence T proves $\text{RB} \rightarrow \text{Con}(T)$. If $T + \neg \text{RB}$ were inconsistent, the deduction theorem would give $T \vdash \text{RB}$ and hence $T \vdash \text{Con}(T)$, contrary to Gödel's second incompleteness theorem when T is consistent. Thus $T + \neg \text{RB}$ is consistent.

Now work in a model of $T + \neg\text{RB}$. Fix $n \geq 2$. The scheme VP includes $\text{VP}(\Pi_{n+1})$. Mohammad proves that if $\text{VP}(\Pi_{n+1})$ holds while the $C^{(n)}$ -extendible cardinals are bounded, then rank-Berkeley cardinals are unbounded [16, Proposition 6.5 and Theorem 6.7]. Thus the $C^{(n)}$ -extendible cardinals are unbounded. For $n = 1$, boundedness of the $C^{(1)}$ -extendibles implies boundedness of the $C^{(2)}$ -extendibles by Lemma 5.3; the $n = 2$ case therefore gives unboundedly many $C^{(1)}$ -extendibles as well. Hence UE holds. $\square$

Local restoration of Choice. The next definitions record finite bounds on the Lévy complexity needed for reflection. Fix a Gödel numbering of formulas and a primitive-recursive prenex-normal-form operation. For a concrete complexity bound, expand defined predicates and terms using their fixed graph formulas, rename bound variables to avoid capture, replace bounded quantifiers by unbounded quantifiers with membership guards, and eliminate implication and biconditional. If $q(e)$ counts all quantifier occurrences in the resulting formula, put $\ell(e) = q(e) + 1$. Prenex conversion has at most $q(e)$ alternating blocks; a vacuous leading block of either polarity bounds both the formula and its negation by $\ell(e)$. This is a conservative primitive-recursive bound and uses no axioms of the rank structures.

Let $\mathcal{D}$ be a finite list of formulas defining additional predicates, and write $e^{\mathcal{D}}$ for the result of eliminating those predicates. Let Σ be a finite set of codes for formulas whose free variables occur among the fixed ordered tuple

$$(\eta, \delta, \theta, \rho, \alpha, x).$$

Let $\Sigma^{\pm}$ be the closure of Σ under subformulas and negation. Its associated complexity bound is

$$c_{\mathcal{D}}(\Sigma) = 2 + \max\left(\{0\} \cup \{\ell(e^{\mathcal{D}}) : e \in \Sigma^{\pm}\}\right).$$

Thus two ranks in $C^{(c_{\mathcal{D}}(\Sigma))}$ agree on all formulas in Σ after interpreting the predicates defined by $\mathcal{D}$.

Let $\chi_{\mathcal{D}, \Sigma}(\eta, \delta, \theta, \rho, \alpha, x)$ be the first-order formula

$$\exists \bar{\delta} \exists \bar{\theta} \exists \bar{\rho} \exists \bar{\alpha} \exists \bar{x} \exists J \Phi_{\mathcal{D}, \Sigma},$$

where $\Phi_{\mathcal{D}, \Sigma}$ is the conjunction of

$$\eta < \bar{\delta} < \bar{\theta} < \bar{\theta} + \omega < \bar{\rho} < \delta < \theta < \theta + \omega < \rho, \quad \bar{\alpha} < \bar{\theta}, \quad \bar{x} \in V_{\bar{\rho}},$$

the assertion that $J : V_{\bar{\rho}} \rightarrow V_{\rho}$ is elementary with respect to the canonical satisfaction relations, the equations

$$\text{crit}(J) = \bar{\delta}, \quad J(\bar{\delta}) = \delta, \quad J(\bar{\theta}) = \theta, \quad J(\bar{\alpha}) = \alpha, \quad J(\bar{x}) = x,$$

the assertion $\bar{\rho} \in C^{(c_{\mathcal{D}}(\Sigma))}$, and the literal finite conjunction

$$\bigwedge_{e \in \Sigma} e^{\mathcal{D}}[\eta, \bar{\delta}, \bar{\theta}, \bar{\rho}, \bar{\alpha}, \bar{x}],$$

where the brackets denote primitive-recursive, capture-avoiding simultaneous substitution for the fixed tuple of free variables. Predicate elimination, simultaneous substitution, finite conjunction, rank membership, and set-coded satisfaction are effective operations. Hence the code of $\chi_{\mathcal{D}, \Sigma}$ is primitive recursive in $\mathcal{D}$ and Σ . We use the following second complexity bound:

$$s_{\mathcal{D}}(\Sigma) = 2 + \max\{c_{\mathcal{D}}(\Sigma), \ell(\ulcorner \chi_{\mathcal{D}, \Sigma} \urcorner)\}.$$

We write $V_{\gamma} \prec_{\Sigma_1^*} V$ for Spoerl's strengthened form of Σ_1 -elementarity [19, Definition 20]. We use Woodin's choiceless definition of supercompactness: a cardinal δ is supercompact if, whenever $\gamma > \delta$, $V_{\gamma} \prec_{\Sigma_1^*} V$, and $a \in V_{\gamma}$, there are $\bar{\gamma} < \delta$, $\bar{\delta}$, $\bar{a} \in V_{\bar{\gamma}}$, and an elementary embedding $j : V_{\bar{\gamma}+1} \rightarrow V_{\gamma+1}$ such that $\text{crit}(j) = \bar{\delta}$, $j(\bar{\delta}) = \delta$, $j(\bar{a}) = a$, and $V_{\bar{\gamma}} \prec_{\Sigma_1^*} V$ [19, Definition 21].

For an ordinal ξ , let DC_{ξ} denote dependent choice for histories of length ξ . We use Woodin's collapse

$$C_{\nu}^{\kappa} = \{p : \text{dom}(p) \subseteq \nu \times \kappa, |\text{dom}(p)| < \nu, p(i, \zeta) \in V_{1+\zeta}\},$$

whose conditions are partial functions, ordered by reverse inclusion. Here ν is an infinite regular cardinal; ranks and the lower index are interpreted in the universe preceding that collapse.

Use the following consistent version of the iteration in [22, pp. 323–324]. Let μ be the least failure of dependent choice, start with the trivial $\mathbb{Q}_0$ and $\kappa_0 = \mu$, and at each successor append $C_{\kappa_\alpha}^{\kappa_{\alpha+1}}$, where $\kappa_{\alpha+1}$ is the least successful ground strongly inaccessible cutoff. At a limit take the direct limit if the supremum of the earlier cutoffs is ground strongly inaccessible. Otherwise first take the raw inverse limit and then append the least successful restoration collapse. Its lower index is the raw extension's own Hartogs successor of that supremum. A completed prefix includes this appended collapse. A stage γ is *marked* when $\kappa_\gamma = \gamma$.

Fix the bounded sparse presentation with globally valid normalized coordinate names. Guard a locally valid name by its condition, with the empty name elsewhere, and normalize by taking the union of all globally equivalent names of the least possible rank. This is a specified representative, uses no choice of representatives, and fixes the empty name. Conditions omit empty coordinates; projections are restrictions and inclusions add empty tails. At direct cuts supports are bounded, and inverse limits retain the support requirements at all earlier direct cuts.

Let $a_\eta = \langle s_\eta, K_\eta \rangle$ be the actual complete code through η , including the seed. The six fields of s_η are the carriers, orders, tops, projections, empty-tail inclusions, and prefix-replacement maps; K_η is the cutoff table. The preceding convention defines the name pools from these data. At a marked η , every condition in every earlier row is in V_η , so finite pairing gives $a_\eta \in V_{\eta+\omega}$.

Theorem 5.5 (Woodin's construction and a restricted lift). *Assume that AC fails, let μ be least such that DC_μ fails, and let $\Lambda > \mu$ be supercompact in Woodin's choiceless sense. The specified iteration is defined through Λ . Its endpoint $\mathbb{Q}_\Lambda$ is weakly homogeneous and uniformly Σ_3 -definable over V_Λ . If $G \subseteq \mathbb{Q}_\Lambda$ is generic, write $W = V[G]$ and let G_α be its induced completed-prefix generic.*

- (1) $W_\Lambda \models \text{ZFC}$.
- (2) Every marked $\gamma \leq \Lambda$ is a direct stage, remains strongly inaccessible in W , and satisfies

$$V_\gamma[G_\gamma] = (V[G_\gamma])_\gamma = W_\gamma \models \text{ZFC}.$$

The iteration factors over its completed prefixes by the canonical complete projections. Every ground supercompact $\gamma \leq \Lambda$ is marked. In $V[G_\alpha]$, a later quotient has a lower bound in its separative preorder for every directed family enumerated by an ordinal below κ_α . It preserves $\text{DC}_{<\kappa_\alpha}$ and adds no shorter sequences of old sets. If $t \in \mathbb{Q}_\Lambda$ has α -prefix in G_α , there is a generic G^ with*

$$t \in G^*, \quad G_\alpha^* = G_\alpha, \quad V[G^*] = W.$$

- (3) Suppose $J : V_{\bar{\gamma}+\omega} \rightarrow V_{\gamma+\omega}$ is elementary, $J(\bar{\gamma}) = \gamma < \Lambda$,

$$\text{crit}(J) = \bar{\delta} < \bar{\gamma} < J(\bar{\delta}) = \delta, \quad \kappa_\gamma = \gamma, \quad J(a_{\bar{\gamma}}) = a_\gamma.$$

Then $\bar{\gamma}, \bar{\delta}, \delta$ are marked, and in W_Λ there is an elementary embedding

$$j : W_{\bar{\gamma}} \longrightarrow W_\gamma$$

with $\text{crit}(j) = \bar{\delta}$, $j(\bar{\delta}) = \delta$, and $j(x) = J(x)$ for every ground $x \in V_{\bar{\gamma}}$.

Proof. We give the construction and rank details needed for the restricted lift. The large-cardinal input is the small-embedding characterization in [22, Lemma 222]. Its critical points, and the supercompact cardinals themselves, are strongly inaccessible in the following choiceless sense: every map $V_\xi \rightarrow \kappa$, $\xi < \kappa$, has bounded range in κ . A contrary map at a critical point would be fixed on its domain and values while its image must be cofinal above that critical point. The same characterization, applied to Σ_2 witnesses at sufficiently correct target ranks, gives $V_\kappa \prec_{\Sigma_2} V$ for supercompact κ . Strong inaccessibility gives $V_\kappa \models \text{ZF}$ by bounding the output ranks in Replacement. A least failure of dependent choice is regular: at a singular least failure, dependent choice builds cofinally many coherent shorter branches, whose union is a full branch. Failure of Choice and failure of dependent choice are Σ_2 , so this least failure is below every supercompact.

We first record the strict restoration step, including its use after a small prefix. If $P, \leq_P, 1_P \in V_\Delta$, Δ is ground supercompact, and

$$P \Vdash \nu < \Delta \text{ is infinite regular and } \text{DC}_{<\nu},$$

then, above any prescribed bound below Δ , there is a ground strongly inaccessible $\lambda < \Delta$ such that

$$P * \dot{C}_\nu^\lambda \Vdash \text{Reg}(\lambda) \wedge \text{DC}_{<\lambda}.$$

For the non-strict upper bound Δ , use the proof of [22, Lemma 225] with a ground small embedding fixing P, ν and its order. It captures a name for the serial relation. The source collapse is the restriction below the critical point. Short histories in the smaller extension stay there by closure and $\text{DC}_{<\nu}$; the full collapse well-orders a bounded pool of names for the source relation. Choosing its least admissible successor constructs a branch of length ν without DC_ν . Applying the lift pointwise gives the required branch in the target. The upper inaccessible stays regular: close a small column subcollapse under least bounds for deciding a proposed short cofinal map; strong inaccessibility bounds the possible decisions. All ordinals below the upper cutoff then have cardinality at most ν . Ordinal induction, using coherent cofinal blocks at limits, gives $\text{DC}_{<\Delta}$.

To make the cutoff strict, take a further small embedding with critical point above the prescribed bound and the prefix data. Dependent choice is absolute between a small forcing extension and its Σ_1^* -correct rank extension. Here the extra reflection is used as follows. A failure has a certificate in a transitive ZF set model closed under the required short functions. Such closed models exist by ground supercompactness. Their closure survives the small forcing: for each condition and argument, use the least-rank union of all names forced to give the same value, and collect the resulting uniquely specified function. A supplied closed model, its satisfaction relation and the bounded forcing certificate express the failure by a closed-set assertion reflected by Σ_1^* correctness. Thus a genuine failure reflects to the rank extension, and conversely all relevant short histories of its sets lie in that rank. Applying this equivalence to the lifted small embedding reflects restoration at Δ to restoration at its critical point. Regularity of this smaller upper cutoff follows from the same decision-bound argument. This proves strict restoration without assuming that P preserves supercompactness.

For the quotient bound, fix a completed base $B = \mathbb{Q}_\alpha$, a later stage, and a base name $\dot{f}$ for a directed family with enumeration length $\nu < \kappa_\alpha$. Choose p forcing its type. List all ground triples (d, ξ, q) for which

$$d \leq p, \quad q \Vdash \alpha, \quad d \Vdash_B \dot{f}(\xi) = \check{q}.$$

For each (d, ξ) there is at most one q . Construct a bound in the ground by recursion, retaining the invariant that, whenever u strengthens its prefix through i and $r \leq u \restriction \alpha, d$, replacement of that prefix by r gives $u[r] \leq q \restriction i$.

At a collapse coordinate take the union of the selected tail names, guarded by d , the constructed prefix bound and the selected condition's prefix. In any generic containing the constructed prefix, the invariant puts every selected original prefix into that generic. The union therefore evaluates to the union of the selected collapse graphs. These graphs are compatible: separative comparison projects through prefix replacement, and a contrary evaluated tail extension would lift to a contrary two-step condition. A canonically enumerated union of fewer than the regular lower-index number of domains is still small. The ranks of the possible names are indexed by the low-rank set $B \times \nu$, so ground strong inaccessibility bounds them below the upper cutoff. Normalization preserves the forced comparisons. At each inaccessible direct cut the least support bounds are indexed by the same set, so all possible supports have a common bound below that cut. At inverse cuts take the coherent ground thread. The invariant uses one and the same base strengthening at every coordinate. At the desired endpoint it says that every extension of the constructed condition is compatible with every member of the family in the quotient. This is the required separative lower bound.

These bounds also give the raw inverse-limit assertion used in the construction. Put $\lambda = \sup_{\xi < \beta} \kappa_\xi$ at a non-inaccessible limit. The earlier completed quotients preserve their $\text{DC}_{<\kappa_\xi}$ and regular cardinals, so the raw extension satisfies $\text{DC}_{<\lambda}$ and λ remains a cardinal. Failure of

ground strong inaccessibility supplies an unbounded $V_\rho \rightarrow \lambda$. An earlier collapse enumerates its domain by an ordinal below λ , making λ singular in the raw extension. Dependent choice on coherent cofinal blocks gives DC_λ , hence $\text{DC}_{<\lambda^+}$ since a least failure is regular. It also makes λ^+ regular: a short cofinal sequence and the choice of injections of its entries into λ would inject λ^+ into λ . This proves the lower-index hypotheses before the completion collapse is appended.

At successors strict restoration now supplies the least cutoff. At limits below Λ , ground strong inaccessibility bounds the earlier cutoffs and the raw forcing rank below Λ . Use the direct rule at inaccessible suprema and the preceding raw argument followed by strict restoration otherwise. This simultaneous induction supplies every stage, its complete projections, regular cutoff and dependent choice, with $\alpha \leq \kappa_\alpha < \Lambda$ for $\alpha < \Lambda$. The quotient-bound recursion applies equally at completed successors and at the final direct row. Repeating the induction with any ground supercompact $\gamma \leq \Lambda$ as outer bound gives $\sup_{\alpha < \gamma} \kappa_\alpha = \gamma$. Thus each such γ is marked.

Weak homogeneity has a specified construction. For two collapse conditions, swap the rows used by the first with the first equally many unused rows outside both supports. The moved condition is compatible with the second. At each coordinate transport this operation by the earlier automorphism and normalize its name. Inverse row permutations give its inverse. Empty coordinates stay empty, and common-extension supports lie in the union of the two original supports, preserving all direct cuts. Starting the recursion above a common completed prefix fixes that prefix pointwise. Given t whose prefix lies in the generic, conditions that some such ground automorphism sends below t are dense below that prefix. Meeting this set gives G^* with $t \in G^*$, the same prefix generic and the same full extension. This also makes each raw inverse-limit successor cardinal a constant ordinal, as required by the recursion.

For definability use bounded hierarchy, subname and forcing certificates for these fixed presentation operations. Their graphs have Δ_2 definitions. The assertion $\text{DC}_{<\kappa}$, and its set-forcing version at a supplied presentation, have Π_2 definitions. A least successful cutoff is consequently described by its positive Π_2 test and the Σ_2 failures at all smaller candidates. A supplied entire prefix therefore has a Σ_3 definition. At V_Λ , C_2 correctness identifies the positive and negative tests with their ground meanings. The bounded operations and uniqueness of the recursion identify its actual prefixes. A sparse endpoint condition belongs to some such prefix, giving the uniform Σ_3 definition of the endpoint.

We prove the rank assertion for the actual stages. Put $M_i = V[G_i]$. The induction invariant is

$$M_i \models \forall \rho < i \exists \nu < \kappa_i \exists f : \nu \twoheadrightarrow V_\rho.$$

It is vacuous at zero. Separative closure and DC_ν decide a name for a sequence of length $\nu < \kappa_i$ by a descending history and a final bound. Thus the tail adds no such sequence of old sets. An old short enumeration of X then prevents new subsets of X , by taking inverse images of characteristic functions. At a successor the invariant therefore preserves all ranks through i , and the actual next collapse enumerates them by $\kappa_i < \kappa_{i+1}$. At a limit θ , for $\rho < \theta$ use the specified earlier stage $i = \rho + 1$. Its enumeration still enumerates the same rank after the completed quotient, and its length is below $\kappa_i < \kappa_\theta$. This also proves the assertion at the separately constructed terminal row Λ .

At a marked γ , these enumerations and regularity bound every function from a smaller extension rank into γ . They give Replacement in $(M_\gamma)_\gamma$ by bounding output ranks, and Choice by choosing least enumeration indices. The remaining ZF axioms follow from transitivity and finite rank bounds. The later quotient preserves all ranks through γ and its regularity, so these assertions hold in W . Every member of W_γ has a name in V_γ : place it in an earlier stabilized prefix and use the name pools

$$B_0 = \emptyset, \quad B_{\xi+1} = \mathcal{P}(B_\xi \times P), \quad B_\lambda = \bigcup_{\xi < \lambda} B_\xi.$$

Their evaluations cover the corresponding ranks by the all-witness atomic-forcing name construction. For each fixed $\xi < \gamma$ their rank is below $\zeta + \omega \cdot (\xi + 1) < \gamma$ for a suitable $\zeta < \gamma$. Conversely a name in V_γ evaluates to a set of rank below γ . This proves (1) and the rank equalities in (2).

Finally assume (3), and write $e = \bar{\delta}$. The captured cutoff table gives $\kappa_{\bar{\gamma}} = \bar{\gamma}$. For $i < e$, covariance of the actual cutoff tables gives $J(\kappa_i) = \kappa_i$. If $\kappa_i \geq e$, then $\kappa_i \geq J(e) = \delta > \bar{\gamma}$, a contradiction. Thus $\kappa_i < e$ for every $i < e$. The critical-point argument makes e strongly inaccessible; $\sup_{i < e} \kappa_i = e$, so e and its image δ are marked. Also $\delta < \gamma$. Every individual condition of $\mathbb{Q}_e$ belongs to V_e and is fixed. For $q \in G_{\bar{\gamma}}$,

$$J(q) \restriction \delta = J(q \restriction e) = q \restriction e \in G_\delta.$$

Thus $D = J''G_{\bar{\gamma}}$ is directed in $\mathbb{Q}_\gamma/G_\delta$ and belongs to $V[G_\delta]$. The source generic has rank below $\bar{\gamma} + 2 < \delta$. The proved enumeration invariant at δ gives it an ordinal enumeration of length below δ ; applying J pointwise gives one for D . Take its separative quotient bound t and use (2) at δ . Then G^* has the same full extension, keeps $G_{\bar{\gamma}}$ fixed, and satisfies

$$J''G_{\bar{\gamma}} \subseteq G_\gamma^*.$$

Indeed, a generic containing a separative lower bound of q meets the dense set of conditions below q or incompatible with q , and the incompatible alternative is impossible.

For $\eta = \bar{\gamma}, \gamma$, let N_η be the set of $\mathbb{Q}_\eta$ -names in V_η . These subname-closed sets are mapped correctly by J . The atomic forcing tables on N_η , followed by the negation, conjunction and existential-density clauses with witnesses in N_η , give a forcing table for the set of their evaluations. Finite assignments are finite functions in V_η , so the whole table and its atomic certificates belong to $V_{\eta+\omega}$. Their defining checks are bounded in the supplied sets. Ambient ZF proves existence and uniqueness, and bounded absoluteness identifies their meaning in these larger ranks; no ZF assumption about $V_{\eta+\omega}$ is used. Elementarity and uniqueness therefore make J carry the source table to the target table.

The usual atomic truth proof, and the decision and witness dense sets for the compound clauses, give the truth lemma. Consequently

$$j(\tau^{G_{\bar{\gamma}}}) = J(\tau)^{G_\gamma^*} \quad (\tau \in N_{\bar{\gamma}})$$

is well-defined and elementary. Transfer forcing witnesses for both a formula and its negation to obtain agreement in both directions. The rank equalities already proved identify its domain and target with $W_{\bar{\gamma}}, W_\gamma$. Check-name covariance gives agreement with J on ground elements, including e and every smaller ordinal. Replacement on $N_{\bar{\gamma}}$ forms the graph, whose rank is below $\gamma + \omega < \Lambda$.

For internal coded elementarity, use the set evaluation graphs on names and finite assignments to express the truth lemma as one predicate of the internal formula code in the quotient. The constructor arguments give definable induction on that syntax, hence truth for all internal codes. This avoids external induction on possibly nonstandard formulas and completes the stated lift. $\square$

Let $\mathcal{D}_W$ be a fixed finite list of formulas defining this iteration, its completed prefixes and raw limits, orders and tops, cutoff table, bounded normalized name pools, projections, empty-tail inclusions, prefix-replacement maps, the complete codes a_η , and marked stages.

For each γ , let $\text{Def}(V_\gamma)$ be the collection of subsets of V_γ definable over $(V_\gamma, \in)$ with parameters from V_γ .

Lemma 5.6. *If γ is supercompact in Woodin's choiceless sense, then $V_\gamma \models \text{ZF}$.*

Proof. By [19, Definition 19 and the proof of Theorem 44], no map $V_\beta \rightarrow \gamma$ with $\beta < \gamma$ has unbounded range. If $a \in V_\gamma$, choose $\beta < \gamma$ with $a \in V_\beta$; any cofinal map $a \rightarrow \gamma$ would extend by a constant value to V_β . Lemma 3.1 now gives $V_\gamma \models \text{ZF}$. $\square$

Lemma 5.7. *For every $k \geq 1$ there is an effectively obtained first-order formula $\vartheta_k(p, e, \vec{\tau})$, independent of the endpoint, with the following property. At any marked strongly inaccessible endpoint γ where V_γ computes the actual prefixes, ϑ_k defines over V_γ the forcing relation for $\mathbb{Q}_\gamma$ for formulas of Lévy complexity at most k , with both polarities included. Its truth lemma holds for every $\mathbb{Q}_\gamma$ -generic over $(V_\gamma, \text{Def}(V_\gamma))$. Every ground supercompact endpoint has the required prefix agreement.*

Proof. Use the sparse embeddings to regard the completed prefixes as nested. At a marked endpoint,

$$\mathbb{Q}_\gamma = \bigcup_{\alpha < \gamma} \mathbb{Q}_\alpha, \quad \mathbb{Q}_\alpha, \leq_\alpha \in V_\gamma,$$

and the projections are complete. Every name $\tau \in V_\gamma$ uses a bounded prefix. Indeed its transitive syntactic closure belongs to V_γ , and the least prefix indices of its conditions form a ground function with bounded range, by strong inaccessibility. The endpoint rank theorem says that these low names represent exactly W_γ .

For a bounded formula and given low names and a condition, compute ordinary set forcing at any prefix containing those data. Completeness and bounded absoluteness between transitive extensions make the answer independent of the prefix. The uniform bounded-formula forcing predicate, including its subname and finite evaluation certificates, is first-order over V_γ . The auxiliary sets fit there by the bounded name ranks.

Extend this predicate using the usual negation and conjunction clauses and the existential density clause

$$p \Vdash \exists x \psi(x, \vec{\tau}) \longleftrightarrow \forall q \leq p \exists r \leq q \exists \sigma (\sigma \text{ is a low name} \wedge r \Vdash \psi(\sigma, \vec{\tau})).$$

These are first-order formulas after substituting the fixed iteration definitions. Conditions here range over the definable endpoint forcing; they must be counted as unbounded quantifiers in the final complexity calculation. Induction proves the truth lemma. Existential witnesses have low names, and the required decision and witness classes are definable and dense. Every ambient generic meets them, because a class definable over the set V_γ is externally a ground set.

The resulting low-name structure satisfies ZFC even for a generic only over $(V_\gamma, \text{Def}(V_\gamma))$. To see this, form externally the set name

$$\dot{U} = \{(\tau, 1_{\mathbb{Q}_\gamma}) : \tau \in V_\gamma \text{ is a } \mathbb{Q}_\gamma\text{-name}\}.$$

For each fixed formula, its ordinary set-forcing relation with all quantifiers relativized to $\dot{U}$ agrees with the preceding low-name density recursion. The marked-rank theorem says that the top forces each fixed standard ZFC axiom in $\dot{U}$. Its corresponding low-name forcing assertion is definable over V_γ , so the class-generic truth lemma yields that axiom in the low-name structure. No extension of the given filter to an ambient generic is assumed.

For a fixed Lévy level k , apply the fixed-formula construction to the one partial satisfaction formula Sat_k . Use the check name of the formula code and the canonical name of the finite tuple of parameters. The preceding ZFC conclusion justifies interpreting Sat_k in the low-name structure, so this gives one uniform formula ϑ_k with the required truth clause for every valid code at that level. No definition for formulas of arbitrary complexity at once is used.

Finally, a ground supercompact γ is C_2 -correct. The canonical presentation operations and the positive and negative restoration tests agree at V_γ , so uniqueness of the recursion identifies its internal prefixes with the actual ones. $\square$

For definiteness, fix the following unpacked presentation of the formula from Lemma 5.7. Let $T(\psi)$ be the low-name forcing translation constructed in its proof, with every graph formula from $\mathcal{D}_W$ substituted. Let $\text{Sat}_k^\Sigma(m, e, b)$ be the fixed partial truth formula for Σ_k codes, with arity m , formula code e , and assignment b . For $k \geq 1$ take

$$\begin{aligned} \vartheta_k(t, p, m, e, b) = & (t = p \wedge T(\text{Sat}_k^\Sigma)(p, m, e, b)) \\ & \vee (t \neq p \wedge T(\neg \text{Sat}_k^\Sigma)(p, m, e, b)). \end{aligned}$$

The extra parameter selects the polarity; a tag different from p exists in every endpoint. In the truth clause, m, e are represented by check names and b by the canonical tuple name. This is a formula in the membership language, so its dictionary predicates have already been eliminated. Fix these five free variables without adding a tuple-decoding formula, and put

$$r_k = c_{\mathcal{D}_W}(\{\vartheta_k\}) = q(\vartheta_k) + 3.$$

The equality follows because negation does not change the quantifier count and every subformula has count at most that of the whole formula. Thus this bound refers to the displayed forcing formula itself, with both polarities, and uses the positive index k of the required Σ_k window.

Lemma 5.8. *Suppose $\theta < \Lambda$ are supercompact in Woodin's choiceless sense, Woodin's iteration is defined through Λ , and $\theta, \Lambda \in C^{(r_k)}$. If $G \subseteq \mathbb{Q}_\Lambda$ is generic, then*

$$(V[G])_\theta \prec_{\Sigma_k} (V[G])_\Lambda.$$

Proof. The endpoints are marked and compute their actual prefixes. Write G_θ for the induced prefix generic. Every parameter in W_θ has a name in V_θ . By the choice of r_k and predicate elimination, for each relevant formula φ , tuple of low names $\vec{\tau}$, and $p \in \mathbb{Q}_\theta$, the forcing assertions

$$p \Vdash_{\mathbb{Q}_\theta} \varphi(\vec{\tau}) \quad \text{and} \quad p \Vdash_{\mathbb{Q}_\Lambda} \varphi(\vec{\tau})$$

are equivalent, and the same holds for $\neg\varphi$. If W_θ satisfies $\varphi(\vec{\tau}^{G_\theta})$, the truth lemma supplies a witnessing $p \in G_\theta$. Transfer its forcing assertion and apply soundness to obtain the same truth in W_Λ . Applying the identical argument to $\neg\varphi$ proves agreement in both directions, hence the required Σ_k -elementarity. $\square$

Lemma 5.9. *Let $S \geq s_{\mathcal{D}}(\Sigma)$, suppose $E_S(\delta)$, and let $\eta < \delta < \theta < \theta + \omega < \rho$, with $\rho \in C^{(c_{\mathcal{D}}(\Sigma))}$. If $\alpha < \theta$, $x \in V_\rho$, and*

$$V \models e^{\mathcal{D}}[\eta, \delta, \theta, \rho, \alpha, x] \quad \text{for every } e \in \Sigma,$$

then $\chi_{\mathcal{D}, \Sigma}(\eta, \delta, \theta, \rho, \alpha, x)$ holds.

Proof. Choose $\mu \in C^{(S)}$ above $\rho + \omega$ and let $k : V_\mu \rightarrow V_\nu$ witness $E_S(\delta)$. Thus $\nu \in C^{(S)}$, $\text{crit}(k) = \delta$, and $k(\delta) > \mu > \rho$. The graph of $k \upharpoonright V_\rho$ has rank below $k(\rho) + \omega = k(\rho + \omega) < \nu$, and Lemma 2.1 shows that this restriction is elementary from V_ρ to $V_{k(\rho)}$. Use the following tuple:

$$(\bar{\delta}, \bar{\theta}, \bar{\rho}, \bar{\alpha}, \bar{x}, J) = (\delta, \theta, \rho, \alpha, x, k \upharpoonright V_\rho).$$

Thus $V_\nu \models \chi_{\mathcal{D}, \Sigma}(\eta, k(\delta), k(\theta), k(\rho), k(\alpha), k(x))$. Here $\rho < k(\delta)$ and $k(\eta) = \eta$. By assumption, the original tuple satisfies the conjunction over Σ , and the $C^{(S)}$ -correctness of V_ν makes the same conjunction true there. Elementarity of k pulls the assertion back to V_μ . Since $\mu \in C^{(S)}$ and the formula and its negation have Lévy complexity below S , the reflected assertion is true in V . $\square$

Lemma 5.10. *Every E_s -cardinal is Σ_{s+2} -correct. Moreover, there is a fixed $d_W < \omega$ such that every E_s -cardinal is supercompact in Woodin's sense whenever $s \geq d_W$.*

Proof. An E_s -cardinal is s -choiceless extendible in the sense of Mohammad: the ordinary witness has critical point $\delta > \xi$ for every prescribed $\xi < \delta$. Its Σ_{s+2} -correctness is therefore [16, Proposition 2.2].

Fix formulas $\sigma_0(\xi)$ and $\sigma_1(\delta, \gamma, a)$ expressing $V_\xi \prec_{\Sigma_1^*} V$ and the small-embedding clause in Spoerl's Definition 21, and choose d_W above the Lévy complexities of these formulas and their negations. Let $s \geq d_W$ and assume $E_s(\delta)$. Given $\gamma > \delta$ with $V_\gamma \prec_{\Sigma_1^*} V$ and $a \in V_\gamma$, choose an E_s -witness $k : V_\mu \rightarrow V_\nu$ with $\mu \in C^{(s)}$ above $\gamma + \omega$. By Lemma 2.1, $k \upharpoonright V_{\gamma+1}$ witnesses in V_ν the required assertion for $k(\gamma), k(\delta), k(a)$. Elementarity reflects this to V_μ , and the choice of d_W makes the reflected map and the Σ_1^* -correctness of its source absolute to V . Thus δ is supercompact in Woodin's sense. $\square$

We now apply Woodin's forcing at an E_s -cardinal for large enough s and use Lemmas 5.9 and 5.10 to obtain a ZFC rank model satisfying any given finite subset of ZFC + VP.

Theorem 5.11. *For every $N \geq 1$ there is an effectively determined integer t_N such that the following holds. Suppose $V \models \text{ZF} + \text{UE} + \neg\text{AC}$, and let μ be least such that DC_μ fails. Let $\Lambda > \mu$ be an E_{t_N} -cardinal, and let $G \subseteq \mathbb{Q}_\Lambda$ be generic for Woodin's forcing. Then $M = (V[G])_\Lambda$ satisfies ZFC and, for every $1 \leq n \leq N$, has a proper class of $C^{(n)}$ -extendible cardinals.*

Proof. For $\gamma \leq \Lambda$, write $M_\gamma = (V[G])_\gamma$ for the rank- γ part of M . Let $k = N + 1$ and put $r = r_k$. Let Ξ_N be the finite set of formulas asserting that $\theta \in C^{(r)}$, that θ is supercompact in Woodin's sense, and that $x = a_\theta$ is the actual complete iteration code defined by $\mathcal{D}_W$. Thus the captured code includes the cutoff table and all six forcing tables. The two complexity bounds needed below are

$$c_N = c_{\mathcal{D}_W}(\Xi_N), \quad s_N = \max\{r, d_W, s_{\mathcal{D}_W}(\Xi_N)\}.$$

Let t_N be four more than the maximum of s_N and the bounds ℓ for $E_{s_N}(\xi)$, the assertion that the E_{s_N} -cardinals are unbounded, membership in $C^{(c_N)}$, and the assertion that $C^{(c_N)}$ is unbounded, after full predicate elimination. These bounds include both polarities. Compute in order r, Ξ_N, c_N , the reflected formula $\chi_{\mathcal{D}_W, \Xi_N}$, s_N , and then t_N . Once the coding is fixed, this is a noncircular primitive-recursive construction in N .

By Lemma 5.10, $\Lambda \in C^{(t_N+2)}$ and Λ is supercompact in Woodin's sense. Theorem 5.5 gives $M \models \text{ZFC}$. Since UE asserts that the E_{s_N} -cardinals are unbounded, the choice of t_N implies that they are unbounded below Λ ; the same argument applies to $C^{(c_N)} \cap \Lambda$.

For each $1 \leq n \leq N$, the Bagaria–Poveda characterization says that, in M , a cardinal δ is $C^{(n)}$ -extendible if and only if, for a proper class of $\theta \in (C^{(n+1)})^M$ and every $\alpha < \theta$, there are $\bar{\delta} < \bar{\theta} < \delta$, $\bar{\alpha} < \bar{\theta}$, and an elementary embedding $e : M_{\bar{\theta}} \rightarrow M_\theta$ such that $\bar{\theta} \in (C^{(n+1)})^M$, $\text{crit}(e) = \bar{\delta}$, $e(\bar{\delta}) = \delta$, and $e(\bar{\alpha}) = \alpha$ [2, Corollary 2.7].

Fix $\eta < \Lambda$ above μ , and choose an E_{s_N} -cardinal δ with $\eta < \delta < \Lambda$. Let $\delta < \theta < \Lambda$ be another such cardinal and let $\alpha < \theta$. Choose $\rho \in C^{(c_N)} \cap \Lambda$ above $\theta + \omega$, and put $x = a_\theta$. The full-code bound gives $x \in V_{\theta+\omega} \subseteq V_\rho$. Lemma 5.10 and the specified recursion show that these objects satisfy Ξ_N . Lemma 5.9 gives

$$\eta < \bar{\delta} < \bar{\theta} < \bar{\theta} + \omega < \bar{\rho} < \delta$$

and $J_0 : V_{\bar{\rho}} \rightarrow V_\rho$ such that

$$\text{crit}(J_0) = \bar{\delta}, \quad J_0(\bar{\delta}) = \delta, \quad J_0(\bar{\theta}) = \theta, \quad J_0(\bar{\alpha}) = \alpha, \quad J_0(\bar{x}) = a_\theta.$$

The reflected instances of Ξ_N give $\bar{\theta} \in C^{(r)}$, show that $\bar{\theta}$ is supercompact in Woodin's sense, and identify $\bar{x}$ with the actual $a_{\bar{\theta}}$. This identification uses the finite correctness bound for the full eliminated code predicate. It is stronger than recovering a source forcing carrier alone.

Set $J = J_0 \upharpoonright V_{\bar{\theta}+\omega}$. Lemma 2.1 shows that $J : V_{\bar{\theta}+\omega} \rightarrow V_{\theta+\omega}$ is elementary.

Since $a_{\bar{\theta}} \in V_{\bar{\theta}+\omega}$, the restriction retains $J(a_{\bar{\theta}}) = a_\theta$. The map J therefore satisfies Theorem 5.5(3), with $\bar{\delta} < \bar{\theta} < J(\bar{\delta}) = \delta < \theta$. It gives directly an elementary $e : M_{\bar{\theta}} \rightarrow M_\theta$ with critical point $\bar{\delta}$, $e(\bar{\delta}) = \delta$ and $e(\bar{\alpha}) = \alpha$. The last equality uses agreement with the ground map on ordinals. The graph belongs to M by the theorem's bound $\text{rank}(e) < \theta + \omega < \Lambda$. The ranks $\bar{\theta}$, θ , and Λ are marked and belong to $C^{(r)}$.

Lemma 5.8, applied twice, gives

$$M_{\bar{\theta}} \prec_{\Sigma_{N+1}} M \quad \text{and} \quad M_\theta \prec_{\Sigma_{N+1}} M.$$

Consequently $\bar{\theta}, \theta \in (C^{(n+1)})^M$ for every $n \leq N$. As in the proof of Lemma 5.2, $M_{\bar{\theta}}$ regards the critical point $\bar{\delta}$ as a cardinal, so M_θ regards $\delta = e(\bar{\delta})$ as a cardinal. Any contrary witness in M already belongs to M_θ , so δ is a cardinal in M .

For the fixed δ , the ordinal θ can be chosen arbitrarily high below Λ , and $\alpha < \theta$ was arbitrary, so the characterization shows that δ is $C^{(n)}$ -extendible for every $n \leq N$. The cardinal δ was arbitrary above η , and η was arbitrary above μ ; hence these cardinals are unbounded below $\Lambda = \text{Ord}^M$, so each class is proper in M . $\square$

Lemma 5.12. *For every finite set F_0 of instances of the arbitrary-language Vopěnka scheme, there is an effectively determined $N \geq 1$ such that*

$$\text{ZFC} + \text{VP}(\Pi_{N+1}) \vdash F_0.$$

Proof. Work in ZFC. Fix a set-sized finitary language L , and let B be the disjoint union of ω and the set of pairs (φ, n) , where φ is an L -formula with free variables among the first n variables. Choose a binary relation T on B whose only endomorphism is the identity [23, Theorem 2]. This one set parameter will be common to every coded L -structure.

For a structure A with domain D , use a fixed finite relational signature with three unary sorts: labels B , elements D , and finite assignments $D^{<\omega}$. Its other relations are the binary relation T on labels, a binary length relation between assignments and natural-number labels, a ternary evaluation relation (t, i, a) saying $t(i) = a$, and a binary satisfaction relation between (φ, n) and assignments of length n . Use disjoint tags and the canonical satisfaction relation for the set structure A . This gives a unique code $\text{Enc}(A)$ by a fixed first-order definition from A, L, T .

An elementary map between two such codes induces an endomorphism of (B, T) , so it fixes every label. Its restriction to the element sort is a map f , and the length and evaluation relations force the image of an assignment t to be $f \circ t$. Preservation and reflection of the satisfaction relation therefore give

$$A \models \varphi[t] \iff A' \models \varphi[f \circ t]$$

for every L -formula. Thus $f : A \rightarrow A'$ is elementary. The code determines A from its element sort and its satisfaction relation on atomic formulas.

For completeness, this fixed finite relational signature can be encoded in one binary relation E and one unary relation R . Use finitely many marker vertices forming a directed path, with R true exactly on them. Take a domain marker, a marker for each relation symbol, and a marker for each argument position. For each true relation tuple add a record vertex and its position vertices. Put edges from the domain marker to the original elements, from the relation marker to the record, from the record to its position vertices, from each position marker to the corresponding position vertex, and from that position vertex to the indicated element. Apart from the marker path these are all the edges. Every marker is definable as a specified finite successor of the first R -vertex. The original domain is therefore definable, and a relation is defined by existence of the appropriate record and its marked position vertices. Relativizing quantifiers to that domain proves, by induction on formulas, that an elementary map of these graph codes restricts to an elementary map of the finite relational codes.

Let Code be this composite canonical coding. It has a definable inverse on its image. For a class defined by $\Phi(A, z, L)$, define its coded class by

$$\exists A(\Phi(A, z, L) \wedge X = \text{Code}(A, L, T)).$$

If the class of codes were a set, Replacement using the decoding would make the original class a set. Thus properness is preserved, and distinct codes decode to distinct structures. The fixed coding definition has finite Lévy complexity, independently of the size of L and the values of z, T . Substitute this definition, calculate a syntactic bound for each of the finitely many input formulas, and take $N \geq 1$ large enough to put them all in Π_{N+1} form. Parameterized $\text{VP}(\Pi_{N+1})$ for the fixed graph signature now gives every required instance. The construction of this bound is effective. $\square$

Theorem 5.13.

$$\text{Con}(\text{ZF} + \text{VP}) \quad \text{if and only if} \quad \text{Con}(\text{ZFC} + \text{VP}).$$

Proof. By Proposition 5.4, take a countable model

$$\mathfrak{M} \models \text{ZF} + \text{VP} + \text{UE} + \text{“there are no rank-Berkeley cardinals”}.$$

Let F be a finite subset of the first-order theory $\text{ZFC} + \text{VP}$. If $\mathfrak{M} \models \text{AC}$, then $\mathfrak{M} \models F$.

Now assume that $\mathfrak{M} \models \neg \text{AC}$. Choose a standard $N \geq 1$ for the finitely many instances of VP in F , as in Lemma 5.12. Choose t_N as in Theorem 5.11 and choose in $\mathfrak{M}$ a sufficiently large E_{t_N} -cardinal Λ . The finite computation of t_N at standard N is correct inside $\mathfrak{M}$, even if its natural numbers are nonstandard.

The countable model $\mathfrak{M}$ need not be externally well-founded. Enumerate externally all of its internal dense subsets of $\mathbb{Q}_\Lambda$ and construct a descending sequence meeting them. Its upward closure is an $\mathfrak{M}$ -generic filter H . Define $\mathfrak{M}[H]$ as the quotient of its internal names, where

$$\tau \sim_H \sigma \iff \exists p \in H \mathfrak{M} \models p \Vdash \tau = \sigma,$$

and define membership by the analogous forced-membership clause. Atomic forcing identities make the quotient well-defined. Induction on each external standard formula gives its truth

lemma; the decision and witness dense sets are internal sets by Separation. No evaluation of names by external well-founded recursion is used.

Let u be the internal set $V_{\check{\lambda}}$ of $\mathfrak{M}[H]$, and let W be its external membership structure. Truth of a standard formula in W is truth of its relativization to u in the quotient. Read Theorem 5.11 as the corresponding set-forcing assertions, one standard axiom or sentence at a time. The truth lemma gives $W \models \text{ZFC}$ and unboundedly many $C^{(n)}$ -extendible cardinals for $1 \leq n \leq N$.

For a parameterized Π_{N+1} class in W , add a vacuous existential quantifier to view its definition as Σ_{N+2} . Choose a $C^{(N)}$ -extendible κ with the type and all parameters in H_{κ}^W . If the class is proper, choose a member outside H_{κ}^W . Bagaria's Theorem 4.11 [1] supplies an elementarily embeddable member inside H_{κ}^W , so the two members are distinct. Thus W satisfies the required parameterized $\text{VP}(\Pi_{N+1})$ scheme. Lemma 5.12 gives every instance of VP in F , and therefore $W \models F$.

Every finite subset of $\text{ZFC} + \text{VP}$ is therefore satisfiable. By first-order compactness, $\text{ZFC} + \text{VP}$ has a model. The converse implication follows by forgetting Choice. $\square$

This proves Theorem B.

Remark 5.14. For a finite set of axioms F , the rank extension used in the proof may depend on F . Compactness therefore proves equiconsistency, but does not give a class-generic $\text{ZFC} + \text{VP}$ extension of the original model. Question 9.2 asks whether every countable model admits one.

6. INDEPENDENCE OF CHOICE PRINCIPLES

Theorem 3.3 applies to the classical symmetric constructions in this section and the next two; Theorem 5.13 gives a ground model of $\text{ZFC} + \text{VP}$ when one is required.

Corollary 6.1.

$$\text{Con}(\text{ZF} + \text{VP}) \text{ implies } \text{Con}(\text{ZF} + \text{VP} + \neg\text{AC} + \neg\text{DC}).$$

Consequently AC is independent of $\text{ZF} + \text{VP}$, relative to $\text{Con}(\text{ZF} + \text{VP})$.

Proof. Let $\mathfrak{M}$ be a countable model of $\text{ZF} + \text{VP}$, and form its basic Cohen symmetric extension. The canonical set $A = \{x_n : n < \omega\}$ of Cohen reals is infinite and Dedekind-finite [12, Section 5.3]. Hence AC and DC fail. Theorem 3.3 preserves VP , while Theorem 5.13 gives the opposite consistency implication for AC . $\square$

Lemma 6.2. Assume $\text{ZFC} + \text{VP}$. There is a set-sized symmetric extension satisfying

$$\text{ZF} + \text{VP} + \text{DC} + \neg\text{AC}.$$

Proof. Let $\mathbb{P}_1$ be the forcing of countable partial functions from $\omega_1 \times \omega_1$ to 2, ordered by reverse inclusion. Let the full permutation group of ω_1 act on the first coordinate, and let the normal filter be generated by pointwise stabilizers of countable subsets of ω_1 . The forcing is σ -closed and the filter is σ -complete, so the symmetric extension satisfies DC by Karagila's preservation theorem [14, Lemma 3.1]. Theorem 3.3 gives $\text{ZF} + \text{VP}$.

For $\xi < \omega_1$, let a_ξ be the ξ th generic subset of ω_1 and put $A = \{a_\xi : \xi < \omega_1\}$. The name for A is invariant, while the name for a_ξ has support $\{\xi\}$, so A belongs to the symmetric extension. Suppose that p forces a hereditarily symmetric name $\dot{f}$, supported by a countable $E \subseteq \omega_1$, to enumerate A . Enlarge E to contain the first-coordinate support of p . Choose $\xi \notin E$, $q \leq p$, and an ordinal γ such that

$$q \Vdash_{\text{sym}} \dot{f}(\check{\gamma}) = \dot{a}_\xi.$$

Choose ζ outside E and outside the countable set of first coordinates occurring in q , and let π transpose ξ and ζ . Then π fixes $\dot{f}$ and p , and

$$\pi q \Vdash_{\text{sym}} \dot{f}(\check{\gamma}) = \dot{a}_\zeta.$$

Since q and πq are compatible, choose a common extension. It forces $\dot{a}_\xi \neq \dot{a}_\zeta$ while forcing both names to equal $\dot{f}(\check{\gamma})$, a contradiction. Thus A is not well-orderable, and AC fails. $\square$

Theorem 6.3.

$$\text{Con}(\text{ZF} + \text{VP}) \text{ implies } \text{Con}(\text{ZF} + \text{VP} + \text{DC} + \neg\text{AC}).$$

Proof. Let $\mathfrak{M} \models \text{ZF} + \text{VP}$ be countable. If $\mathfrak{M} \models \text{AC}$, apply Lemma 6.2. If $\mathfrak{M} \models \neg\text{AC} + \neg\text{DC}$, choose a singular weakly LS cardinal κ by Corollary 4.4. Usuba's forcing $\text{Col}(V_\kappa)$ forces DC [21, Proposition 3.6], while Theorem 3.3 preserves VP; if the extension satisfies AC, apply Lemma 6.2 once more. The remaining case already satisfies $\text{DC} + \neg\text{AC}$. $\square$

Corollary 6.4. *Relative to $\text{Con}(\text{ZF} + \text{VP})$, the axiom DC is independent of*

$$\text{ZF} + \text{VP} + \neg\text{AC}.$$

Proof. Corollary 6.1 gives a model with $\neg\text{DC}$, and Theorem 6.3 gives one with DC, from the same consistency hypothesis. $\square$

Let RR denote the rigid relation principle: every set carries a binary relation with no nonidentity automorphism.

Corollary 6.5. *The following consistency implications hold:*

$$\begin{aligned} \text{Con}(\text{ZF} + \text{VP}) &\text{ implies } \text{Con}(\text{ZF} + \text{VP} + \neg\text{RR}), \\ \text{Con}(\text{ZF} + \text{VP}) &\text{ implies } \text{Con}(\text{ZF} + \text{VP} + \text{RR} + \neg\text{AC} + \neg\text{DC}). \end{aligned}$$

Proof. Hamkins and Palumbo prove that every model of ZF has a set-sized symmetric extension in which RR fails, and that RR holds in the basic Cohen model [8, Theorems 5 and 6]. Apply Theorem 3.3 to the first construction over a model of $\text{ZF} + \text{VP}$. For the second implication, first use Theorem 5.13 to obtain a model of $\text{ZFC} + \text{VP}$, and then use the Cohen model from the proof of Corollary 6.1; it satisfies $\neg\text{AC} + \neg\text{DC}$, and Theorem 3.3 preserves VP. $\square$

7. THE FEFERMAN–LÉVY MODEL

Theorem 7.1. *If $\text{ZF} + \text{VP}$ is consistent, then so is $\text{ZF} + \text{VP} + \neg\text{AC} + \neg\text{DC}$ together with the assertion that $\mathbb{R}$ is a countable union of countable sets.*

Proof. By Theorem 5.13, it suffices to start with $V \models \text{ZFC} + \text{VP}$. Let $\mathbb{P}_0$ be trivial, let N_0 consist of the canonical names for the ground-model reals, and choose an infinite regular cardinal $\kappa_1 > |N_0|$. Recursively, after $\kappa_1, \dots, \kappa_n$ have been chosen, define the finite-stage forcing by

$$\mathbb{P}_n = \prod_{1 \leq m \leq n}^{\text{fin}} \text{Coll}(\omega, \kappa_m),$$

let N_n be the set of nice $\mathbb{P}_n$ -names for reals, and choose an infinite regular cardinal $\kappa_{n+1} > \max\{\kappa_n, |N_n|\}$. Following the standard Feferman–Lévy construction [12, Theorem 10.6], use the full forcing

$$\mathbb{P} = \prod_{1 \leq m < \omega}^{\text{fin}} \text{Coll}(\omega, \kappa_m).$$

Let Γ consist of the permutations acting independently on the ω -coordinate of each column, and let $\mathcal{F}$ be generated by the subgroups H_n fixing the first n columns pointwise, with $H_0 = \Gamma$. Let G be generic and $W = \text{HS}^G$. Proposition 2.2 and Theorem 3.3 give

$$W \models \text{ZF} + \text{VP}.$$

For $n < \omega$, let G_n be the restriction of G to columns $1, \dots, n$, with G_0 trivial. We use the notation

$$R_n = \mathbb{R}^{V[G_n]}.$$

If $x \in W$ is a real, $\dot{x}$ has support H_n , and $p \Vdash_{\text{sym}} \check{k} \in \dot{x}$, the usual support argument gives

$$p \restriction n \Vdash_{\text{sym}} \check{k} \in \dot{x}.$$

If $q \leq p \restriction n$ forced the opposite decision, choose $\pi \in H_n$ moving the finitely many coordinates of p above column n away from those of q . Then πp and q are compatible and π fixes $\dot{x}$, a contradiction. Therefore

$$x = \{k \in \omega : \exists q \in G_n (q \Vdash_{\mathbb{P}} \check{k} \in \dot{x})\},$$

and ordinary and symmetric forcing agree in this atomic formula for these hereditarily symmetric names. Hence $x \in V[G_n]$. Conversely, every $\mathbb{P}_n$ -name is hereditarily fixed by H_n , and hence $V[G_n] \subseteq W$. Consequently,

$$\mathbb{R}^W = \bigcup_{n < \omega} R_n.$$

Regard the set N_n fixed above as a set of $\mathbb{P}$ -names. The names used below are

$$\dot{R}_n = \{\langle \dot{y}, 1_{\mathbb{P}} \rangle : \dot{y} \in N_n\}, \quad \dot{S} = \{\langle \langle \check{n}, \dot{R}_n \rangle^{\bullet}, 1_{\mathbb{P}} \rangle : n < \omega\}.$$

Every automorphism preserves the first n columns as a set and permutes N_n , so $\dot{R}_n$ and $\dot{S}$ are invariant. Their constituent names are supported by H_n , hence $\dot{S} \in \text{HS}$ and

$$\dot{S}^G = \langle R_n : n < \omega \rangle.$$

Hence the entire sequence lies in W .

Finally, each R_n is countable in W . Fix $n < \omega$ and, in V , well-order N_n . Mapping each real in R_n to the least index of a nice name evaluating to it gives, in $V[G_n] \subseteq W$, an injection $e_n : R_n \rightarrow |N_n|^V$, where $|N_n|^V < \kappa_{n+1}$. The next generic column adds a surjection $g_{n+1} : \omega \rightarrow \kappa_{n+1}$. Extending e_n^{-1} by a fixed value off $\text{ran}(e_n)$ and composing with g_{n+1} gives a surjection from ω onto R_n . For $n = 0$, the filter G_0 is trivial, N_0 is the chosen set of canonical names, and g_1 is the first generic column.

Thus W contains the sequence $\langle R_n : n < \omega \rangle$ of countable sets with union $\mathbb{R}^W$. By Cantor's theorem, $\mathbb{R}^W$ is uncountable. Hence countable choice fails, and so does DC. $\square$

8. THE FULL SOLOVAY MODEL

Suppose for this section that $V \models \text{ZFC} + \text{VP}$. By the final clause of Lemma 3.2, fix a measurable cardinal κ and a nonprincipal κ -complete ultrafilter U on κ . Let $G \subseteq \text{Coll}(\omega, < \kappa)$ be V -generic, put $R = (\omega^\omega)^{V[G]}$, and define

$$(5) \quad \text{Sol} = \text{HOD}_{V \cup R}^{V[G]},$$

where individual elements of V and individual members of R are allowed as parameters. We use LM, BP, and PSP for the assertions that every set of reals is, respectively, Lebesgue measurable, has the Baire property, and has the perfect set property.

Lemma 8.1. *The model Sol satisfies $\text{ZF} + \text{DC} + \text{LM} + \text{BP} + \text{PSP}$, and $\kappa = \omega_1^{\text{Sol}}$.*

Proof. The model in (5) is the full Solovay model in the precise convention of [17, Definition 4 and the following paragraph]. Since a measurable cardinal is inaccessible in ZFC, Solovay's argument applies. For the parameter convention, use the intermediate ground-and-real-definable sets in [18, III, Sections 1.3–1.11]: those arguments explicitly allow an arbitrary ground-set parameter. Absorb the real parameter into a bounded collapse extension. Over that ground, the random-real Boolean value gives a Borel approximation to each such set, and its error is contained in the Borel null set of non-random reals. The Cohen/category version gives the Baire property. For an uncountable such set, the construction from pairs of mutually generic collapsing maps gives a continuous injection of 2^ω into it. The Borel and perfect-set codes are reals and therefore belong to Sol ; their interpretations agree with those in $V[G]$.

For the hereditary passage, the full model is closed under countable sequences taken in $V[G]$. Indeed, choose in $V[G]$ defining data for the terms of a sequence. The reals in these data combine into one real. The countable sequences of ground parameters and of ordinal parameters have bounded-stage names: the κ -chain condition bounds the union of their deciding antichains. Such a bounded-stage name and a real coding its generic filter define the whole sequence from allowed parameters. Heredity follows from heredity of its terms. This also transfers a countable enumeration when the set of reals is countable in $V[G]$. A dependent-choice sequence

chosen in the ambient ZFC extension consequently belongs to $\mathcal{Sol}$. Together with the usual generalized-HOD proof of ZF, these facts give $\mathcal{Sol} \models \text{ZF} + \text{DC} + \text{LM} + \text{BP} + \text{PSP}$, as in [18, III, Sections 2.4–2.11].

The collapse makes every $\alpha < \kappa$ countable in $\mathcal{Sol}$. Since κ is inaccessible, $\text{Coll}(\omega, < \kappa)$ is κ -c.c. and preserves the uncountability of κ . Hence $\kappa = \omega_1^{\mathcal{Sol}}$. $\square$

Lemma 8.2. *If $x \in \mathcal{Sol}$ is a set of ordinals, then $x \in V[G_\beta]$ for some $\beta < \kappa$, where $G_\beta = G \cap \text{Coll}(\omega, < \beta)$. Moreover, if $\langle A_\xi : \xi < \delta \rangle \in \mathcal{Sol}$, where $\delta < \kappa$ and every $A_\xi \subseteq \kappa$, then the whole sequence belongs to one such bounded-stage extension.*

Proof. Let $x \in \mathcal{Sol}$ be a set of ordinals, and choose an ordinal θ with $x \subseteq \theta$. There are a formula φ , ordinal parameters $\vec{\gamma}$, a parameter $a \in V$, and a real $r \in V[G]$ such that

$$\alpha \in x \quad \text{if and only if} \quad V[G] \models \varphi(\alpha, \vec{\gamma}, a, r) \quad (\alpha < \theta).$$

A nice name for r has support of size less than κ , and hence bounded support, so $r \in V[G_\beta]$ for some $\beta < \kappa$. Let $\mathbb{P}^\beta$ be the quotient forcing over $V[G_\beta]$. Since $\mathbb{P}^\beta$ is weakly homogeneous and its automorphisms fix the parameters,

$$x = \{\alpha < \theta : 1_{\mathbb{P}^\beta} \Vdash \varphi(\check{\alpha}, \check{\vec{\gamma}}, \check{a}, \check{r})\}.$$

The definability of the forcing relation and Separation in $V[G_\beta]$ therefore give $x \in V[G_\beta]$. This is the localization argument for the full Solovay model [17, Definition 4, Fact 5, and the paragraph following Fact 5].

For the final assertion, code the sequence by

$$C = \{\kappa \cdot \xi + \alpha : \xi < \delta, \alpha \in A_\xi\}.$$

Then $C \in \mathcal{Sol}$, so $C \in V[G_\beta]$ for some $\beta < \kappa$. Since

$$A_\xi = \{\alpha < \kappa : \kappa \cdot \xi + \alpha \in C\},$$

the sequence $\langle A_\xi : \xi < \delta \rangle$ belongs to $V[G_\beta]$. $\square$

Lemma 8.3. *In $\mathcal{Sol}$, the set*

$$U^* = \{A \in \mathcal{P}(\kappa)^{\mathcal{Sol}} : (\exists Y \in U) Y \subseteq A\}$$

is a nonprincipal, internally κ -complete ultrafilter on κ .

Proof. Since $U \in V \subseteq \mathcal{Sol}$, Separation gives $U^* \in \mathcal{Sol}$. For $\beta < \kappa$, define

$$U_\beta = \{X \in \mathcal{P}(\kappa)^{V[G_\beta]} : (\exists Y \in U) Y \subseteq X\}.$$

The forcing $\text{Coll}(\omega, < \beta)$ has ground-model cardinality less than κ , so [3, Theorem 2.5(2)] shows that U_β is a κ -complete ultrafilter. If $A \subseteq \kappa$ belongs to $\mathcal{Sol}$, choose $\beta < \kappa$ with $A \in V[G_\beta]$ by Lemma 8.2. Then $A \in U^*$ if and only if $A \in U_\beta$, so exactly one of A and $\kappa \setminus A$ belongs to U^* .

To verify internal κ -completeness, let $\delta < \kappa$ and $\langle A_\xi : \xi < \delta \rangle \in \mathcal{Sol}$, with every $A_\xi \in U^*$. By Lemma 8.2, the sequence $\langle A_\xi : \xi < \delta \rangle$ belongs to $V[G_\beta]$ for some $\beta < \kappa$. Since each $A_\xi \in U^*$, we have $A_\xi \in U_\beta$. The κ -completeness of U_β therefore gives

$$\bigcap_{\xi < \delta} A_\xi \in U_\beta.$$

The intersection belongs to $\mathcal{Sol}$, and its membership in U_β therefore implies that it belongs to U^* . Finally, no singleton belongs to U^* because U is nonprincipal. Thus U^* is nonprincipal and internally κ -complete. $\square$

Lemma 8.4. *The model $\mathcal{Sol}$ is the symmetric extension arising from a set-sized symmetric system in V .*

Proof. Let $\mathbb{B}$ be the Boolean completion of $\text{Coll}(\omega, < \kappa)$, let $H \subseteq \mathbb{B}$ be the induced generic, and define the Boolean class name below using the condition-first convention of [15]:

$$\dot{R}_{\text{can}} = \{(b, \tau) : \tau \text{ is a } \mathbb{B}\text{-name and } 0 < b \leq \|\tau \in \check{\omega}\|_{\mathbb{B}}\}.$$

Then $\dot{R}_{\text{can}}^H = R$. Equivariance of Boolean values gives $\pi \dot{R}_{\text{can}} = \dot{R}_{\text{can}}$ for every $\pi \in \text{Aut}(\mathbb{B})$. Moreover, $\dot{R}_{\text{can}}$ self-reflects in the sense of [15, Definitions 9.1–9.2]: for any fixed ground-model real r_0 , both $\dot{R}_{\text{can}}$ and $\check{r}_0$ have stabilizer $\text{Aut}(\mathbb{B})$, and $1_{\mathbb{B}} \Vdash \check{r}_0 \in \dot{R}_{\text{can}}$.

Consequently, by [15, Proposition 9.4], the symmetric extension arising from $\mathcal{O}(\mathbb{B}, \dot{R}_{\text{can}})$ is exactly

$$\text{HOD}_{V \cup R}^{V[H]} = \text{Sol},$$

using the displayed self-reflecting name. This symmetric system is set-sized in V : $\mathbb{B}$ and $\text{Aut}(\mathbb{B})$ are sets, and its generated normal filter is a definable subset of $\mathcal{P}(\text{Aut}(\mathbb{B}))$.

In the reverse inclusion in the proof of [15, Proposition 9.4], the comparison generics must generate the same extension, as required by its Lemma 9.3. This restriction is definable from ground-set parameters. Fix in V the set T of canonical Boolean names for subsets of $\mathbb{B}$, and consider only generics H' in $V[H]$ for which every subset of $\mathbb{B}$ in $V[H]$ is $\tau^{H'}$ for some $\tau \in T$. The actual H has this property. Conversely, such an H' has $H \in V[H']$, so $V[H'] = V[H]$. The comparison argument therefore applies without using H as a defining parameter. $\square$

Corollary 8.5. *Con(ZF + VP) implies the consistency of*

ZF + VP + DC + LM + BP + PSP + $\neg$ AC + “ ω_1 carries a nonprincipal ω_1 -complete ultrafilter.”

Proof. By Theorem 5.13, take the starting model to satisfy ZFC + VP. By Lemma 8.4 and Theorem 3.3, $\text{Sol} \models \text{VP}$. Lemma 8.1 gives ZF + DC + LM + BP + PSP and identifies κ with ω_1^{Sol} , while Lemma 8.3 gives the nonprincipal, internally κ -complete ultrafilter. The standard Solovay argument gives $\neg$ AC. $\square$

9. LIMITS OF CHOICE RESTORATION

Question 9.2 asks whether every countable model of ZF + VP has a class-generic extension satisfying ZFC + VP. Under the stated forcing-theorem hypotheses, Proposition 9.1 characterizes when a class forcing preserves $\text{VP}(\Pi_m)$. Proposition 9.4 gives a positive answer under SVC. Corollary 9.6 rules out well-orderable set forcing, while Proposition 9.7 gives an $\aleph_1$ -sized counterexample to the version without countability.

Proposition 9.1. *Let $(M, \mathcal{X}) \models \text{GBC}$ be a countable transitive two-sorted model, where GBC denotes Gödel–Bernays set theory with Global Choice, and let $\mathbb{P} \in \mathcal{X}$ be a class forcing with a largest condition. Assume that the forcing theorem holds for first-order formulas, that the relevant forcing relations belong to $\mathcal{X}$, that a generic may be chosen through every condition, and that every generic extension has the same ordinals and satisfies ZFC. For standard $m \geq 1$, let $K_m(\kappa)$ mean that κ is supercompact if $m = 1$, and $C^{(m-1)}$ -extendible if $m \geq 2$. Then*

$$1_{\mathbb{P}} \Vdash \text{VP}(\Pi_m) \quad \text{if and only if} \quad \forall p \in \mathbb{P} \, \forall \eta \in \text{Ord}^M \, \exists q \leq p \, \exists \kappa > \eta \, q \Vdash K_m(\check{\kappa}).$$

Here the left side means that every coded instance of $\text{VP}(\Pi_m)$ is forced. Consequently, $\mathbb{P}$ forces the full first-order scheme exactly when the right side holds for every standard $m \geq 1$.

Proof. Over ZFC, $\text{VP}(\Pi_m)$ is equivalent to the unboundedness of the cardinals satisfying K_m [1, Theorem 4.3(2), Corollary 4.7, Theorem 4.11, and the paragraph following Theorem 4.12]; arbitrary set-sized languages reduce to the usual coding under Choice [5, Lemma 23]. Suppose first that $\text{VP}(\Pi_m)$ is forced, and fix p and η . Choose a generic G containing p . Some $\kappa > \eta$ in $M[G]$ satisfies K_m , and the preservation of ordinals gives $\kappa \in M$. By the assumed truth lemma and directedness of G , some $q \leq p$ forces $K_m(\check{\kappa})$.

Conversely, for each η , the conditions forcing a K_m -cardinal above η are dense, so every generic extension satisfies $\text{VP}(\Pi_m)$. For each instance, the hypotheses that a generic can be chosen through every condition and that the truth lemma holds imply that the conditions forcing it are dense. Hence $1_{\mathbb{P}}$ forces every instance of $\text{VP}(\Pi_m)$. $\square$

Question 9.2. For every countable first-order model $M \models \text{ZF} + \text{VP}$, do there exist a class part $\mathcal{X}$ over M , a class forcing $\mathbb{P} \in \mathcal{X}$, and an externally $(M, \mathcal{X})$ -generic filter $G \subseteq \mathbb{P}$ such that the quotient-of-names extension satisfies $M[G] \models \text{ZFC} + \text{VP}$?

Question 9.3. For each standard $n \geq 1$, is it relatively consistent with $\text{ZFC} + \text{VP}(\Pi_{n+1})$ that a class forcing, definable in the language of set theory from set parameters, preserves ZFC and $\text{VP}(\Pi_n)$ but destroys $\text{VP}(\Pi_{n+1})$?

Two related results bear on Question 9.3. For $n \geq 2$, suitable Δ_2 -definable class iterations preserve $\text{VP}(\Pi_n)$ over models of $\text{VP}(\Pi_{n+1})$ [2, Theorem 9.6]. On the other hand, Hamkins shows that the first-order scheme need not imply Vopěnka’s principle for arbitrary classes [7, Theorem 11].

Blass introduced the axiom of small violations of Choice, SVC . We use his characterization that, over ZF , SVC is equivalent to the existence of a set forcing which forces AC [4]; see also [14, Section 3].

Proposition 9.4. *Let $M \models \text{ZF} + \text{VP} + \text{SVC}$. There is a set forcing $\mathbb{P} \in M$ such that every M -generic extension by $\mathbb{P}$ satisfies $\text{ZFC} + \text{VP}$. In particular, if M is countable, Question 9.2 has a positive answer for M .*

Proof. By Blass’s characterization, choose $\mathbb{P} \in M$ with $1_{\mathbb{P}} \Vdash \text{AC}$. Set forcing preserves ZF , and the trivial-symmetry case of Theorem 3.3 preserves VP . Thus $1_{\mathbb{P}} \Vdash \text{ZFC} + \text{VP}$. $\square$

Lemma 9.5. *Let $\mathbb{P}$ be a well-orderable set forcing, X a ground-model set, and suppose that for some ordinal κ and name $\dot{f}$,*

$$1_{\mathbb{P}} \Vdash \dot{f} : \check{\kappa} \rightarrow \check{X}.$$

Then X is well-orderable in the ground model.

Proof. Fix a well-order of $\mathbb{P} \times \kappa$. For $x \in X$, let $d(x)$ be the least pair (p, α) such that $p \Vdash \dot{f}(\check{\alpha}) = \check{x}$. Since $1_{\mathbb{P}}$ forces $\dot{f}$ to be surjective, Replacement yields a function $d : X \rightarrow \mathbb{P} \times \kappa$. If $d(x) = d(y) = (p, \alpha)$, then p forces both $\dot{f}(\check{\alpha}) = \check{x}$ and $\dot{f}(\check{\alpha}) = \check{y}$, so $x = y$. Thus d is injective, and hence X is well-orderable. $\square$

Corollary 9.6. *A well-orderable set forcing cannot restore Choice over a model of $\neg\text{AC}$.*

Proof. Let a generic extension by a well-orderable $\mathbb{P}$ satisfy AC , and fix a ground-model set X . In the extension, choose a surjection $f : \kappa \rightarrow X$ from an ordinal κ . By the truth lemma, there are a condition $p \in \mathbb{P}$ and a name $\dot{f}$ such that $p \Vdash \dot{f} : \check{\kappa} \rightarrow \check{X}$. After restricting $\dot{f}$ below p , Lemma 9.5 applied to the well-orderable forcing $\mathbb{P} \restriction p$ shows that X is well-orderable in the ground model. Thus every ground-model set is well-orderable, and the ground model satisfies AC . $\square$

Proposition 9.7. *Assume $\text{Con}(\text{ZF} + \text{VP})$. There is a model N of cardinality $\aleph_1$ satisfying $\text{ZF} + \text{VP} + \neg\text{AC}$ which has no proper generic extension satisfying ZF . Consequently, the analogue of Question 9.2 for all set-sized models is false.*

Proof. By Corollary 6.1, take a countable $M_0 \models \text{ZF} + \text{VP} + \neg\text{AC}$. Enayat’s theorem gives an elementary extension $M_0 \prec N$ of cardinality $\aleph_1$ having no proper end extension to a model of ZF [6, Theorem 5.18]. Elementarity gives $N \models \text{ZF} + \text{VP} + \neg\text{AC}$. The atomic forcing recursion shows that old sets acquire no new members in a quotient-of-names extension. Hence any such extension that adds a set is both proper and an end extension. Therefore no such extension of N can satisfy ZF . $\square$

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